Complex Landscape Rice Extraction Using Integrated Sentinel-2 Spectral–Temporal–Spatial Imagery and a Hybrid Deep Learning Architecture

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***Abstract*— Rice extraction in complex landscapes is a challenging issue in remote sensing, particularly in areas with diverse land-use types and spatiotemporal variability. To enhance the accuracy of rice extraction, this study proposes a novel approach integrating Sentinel-2 spectral–temporal–spatial imagery with a hybrid deep learning architecture for extracting single-cropping and double-cropping rice. First, a time-series dataset of spectral and texture features was constructed to capture the seasonal variations of rice. Second, an active learning strategy was employed to select high-confidence samples, and spectral, temporal, and spatial information was integrated into a unified dataset. Finally, a hybrid deep learning model, convolutional–Transformer hybrid network (CTH-Net), was developed, combining convolutional neural networks (CNNs) and Transformer networks. The model incorporates a fusion module to effectively integrate multiscale temporal, spatial, and spectral features and a residual module to improve gradient flow, mitigating the vanishing gradient problem in deep networks. Results demonstrate that the CTH-Net achieved 99.69% overall accuracy in rice extraction, maintaining >96% accuracy for single-cropping rice, double-cropping rice, and abandoned land. It outperformed models like CNNs, Transformers, long short-term memory (LSTM), and support vector machines (SVMs) in handling fragmented rice distributions and mixed land types, significantly improving extraction accuracy. This study provides an efficient and reliable solution for rice extraction in complex landscapes, supporting agricultural monitoring and management.**

***Index Terms*— Active learning, hybrid deep learning model, rice extraction, Sentinel-2 spectral–temporal–spatial imagery.**

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I. Introduction

RICE is one of the world's primary staple crops, particularly in Asia, where its production and consumption account for a significant portion of global totals [1], [2], [3]. Accurate monitoring of rice cultivation areas is critical to ensuring food security, especially in major producing countries such as China, India, and Indonesia. In these regions, rice is not only a key food source but also a vital component of the agricultural economy [1], [4]. Rice cultivation has distinct irrigation requirements, and this wetland farming practice significantly impacts water usage and greenhouse gas emissions [5], [6], [7]. Therefore, timely and precise information on rice-growing areas is essential for efficient water resource management, addressing climate change, and assessing the environmental impact of agricultural activities. With the continuous advancement of remote sensing technologies, satellite-based rice distribution mapping has become an efficient and reliable method [8], [9], [10], [11]. Remote sensing imagery provides broad coverage of large areas, enabling real-time monitoring of rice-growing regions and their dynamic changes, greatly enhancing the efficiency and accuracy of traditional ground surveys [12], [13]. Open-access satellite imagery, such as from the Sentinel-2 and Landsat series, offers abundant spatiotemporal data. These images enable effective rice extraction in complex landscapes, supporting precision agriculture, land resource management, and informed agricultural policy decision-making [14], [15], [16].

Remote sensing methods for crop mapping can be divided into two categories based on the temporal dimension: single-temporal and multitemporal methods. Single-temporal methods rely on spectral features from remote sensing images at a specific time, offering simple computation and fast processing, which is useful for rapid assessments. However, due to the spatiotemporal variability of rice growth, single-temporal images often fail to capture the full growth dynamics, leading to unstable and inaccurate results. Additionally, adverse weather during critical growth stages can hinder the acquisition of high-quality images, affecting mapping accuracy and timeliness [17]. To address these limitations, multitemporal remote sensing data have been employed to improve crop information extraction. Analyzing multitemporal images throughout the crop growth cycle allows for better use of spectral response

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and growth cycle characteristics, enabling more accurate crop differentiation [18], [19].

Many studies have utilized spectral similarity methods or phenological features for crop extraction from multitemporal data [20], [21], [22], [23], [24], [25], [26], [27]. Although these methods effectively integrate temporal and spectral features, they often overlook spatial characteristics, thereby limiting their applicability for rice extraction in complex landscapes. Spectral features capture vegetation reflectance across multiple bands, enhancing rice separability [8]. However, at certain growth stages, rice shares similar spectral signatures with other vegetation, making classification based solely on spectral information challenging [25]. Spatial features, such as texture, characterize field structure, including planting density and plot shape, aiding in the identification of fragmented rice fields [28]. Nevertheless, spatial information alone is insufficient to distinguish land covers with similar spatial patterns, such as single-cropping and double-cropping rice [20]. Temporal features, which reflect growth dynamics and phenology, are crucial for rice extraction [24]. However, they require frequent observations and are susceptible to cloud cover [27]. Therefore, integrating temporal, spatial, and spectral features leverages complementary information, enhancing classification accuracy and adaptability in complex landscapes. Recent studies have explored temporal–spatial–spectral integration through feature relationship learning-based methods to improve classification accuracy and adaptability [29], [30], [31]. These efforts have shown promise in enhancing crop extraction. However, several challenges remain. First, it remains challenging to fully capture seasonal dynamics and spatial heterogeneity, and existing methods struggle to effectively characterize the interactions between multiscale temporal variations and spatial distribution patterns. Second, more advanced feature fusion strategies are needed to better exploit synergistic relationships among features at different scales.

Multitemporal remote sensing data combined with machine learning techniques, such as random forest (RF) and support vector machines (SVMs), have been applied for crop classification [32], [33], [34]. However, these methods heavily depend on feature engineering to identify relevant features. Additionally, they typically treat multitemporal data as independent variables, overlooking temporal correlations and temporal–spatial dynamics, which limits their ability to capture complex temporal–spatial dependencies effectively [35], [36]. Deep learning has exhibited strong potential in crop mapping [37], [38], [39]. Superior to traditional machine learning methods, it can automatically extract hierarchical features from remote sensing data, reducing manual feature engineering [40], [41]. Its capability to capture temporal correlations and temporal–spatial patterns enables more accurate modeling of complex dependencies [35]. Convolutional neural networks (CNNs) are among the most commonly used architectures, as they can automatically extract deep features related to crop types, geographic location, and spectral information through multilayer convolution, making them widely adopted for crop classification [42], [43], [44]. By applying convolution operations with fixed-size kernels, CNNs efficiently extract local spatial and spectral features [45], [46]. However, their limited receptive fields hinder their ability to model long-range dependencies and global contextual information in time-series data, making it difficult for CNNs to effectively model the temporal–spatial dynamics of rice growth, where both temporal and spatial features are crucial for accurate classification [31], [47]. Given the importance of temporal information, recurrent neural networks (RNNs) are specifically designed for time-series data [48], [49], [50]. In particular, long short-term memory (LSTM) networks introduce gating mechanisms to overcome the gradient vanishing issue in standard RNNs, making them more stable when modeling long-term dependencies [51], [52]. LSTM has been widely used in multitemporal crop classification [53], [54]. However, studies have shown that LSTMs and RNNs may underperform compared to 1-D CNNs for multitemporal Landsat crop classification [55], [56], [57]. Unlike RNNs, Transformers leverage self-attention mechanisms to process entire time-series data in parallel, overcoming the sequential limitations of RNNs and enhancing long-term dependency modeling and computational efficiency [58], [59], [60]. Yuan et al. [61] demonstrated that a Transformer-based model outperforms RNN-based models in crop mapping, offering greater robustness and flexibility for complex temporal–spatial data. Nevertheless, due to their lack of inductive biases, Transformers struggle to capture fine-grained spectral and spatial details, potentially reducing rice extraction accuracy, especially in critical growth stages or complex landscapes [62]. Therefore, integrating the local feature extraction capability of CNNs with the long-range dependency modeling of Transformers is crucial for enhancing rice extraction accuracy and robustness.

Some multinetwork ensemble methods have shown great potential in enhancing crop mapping performance [63], [64], [65]. Most of these methods adopt encoder–decoder structures originally designed for segmentation or sequence tasks, which increases model complexity and limits their suitability for lightweight, pixel-level classification, as pixel-level classification tasks primarily require effective encoders to extract relevant features [66]. Additionally, most models rely on simple combinations of different neural architectures, with less emphasis on in-depth feature interaction mechanisms or targeted fusion strategies. The labor-intensive task of generating training samples constrains the scalability of large-scale crop mapping. To alleviate this, some researchers have developed automatic sample generation methods [67], [68], [69]. Although these methods reduce reliance on manual annotation and improve the scalability of crop mapping, they are not directly applicable to the detailed extraction of single-cropping and double-cropping rice in complex landscapes, highlighting the need for methods to generate large, high-quality training samples for such crops.

To overcome the above-mentioned limitations, this study proposes a lightweight hybrid deep learning architecture that integrates temporal, spatial, and spectral features for pixel-level rice extraction. First, a feature set encompassing spectral, temporal, and spatial information was constructed. Then, high-quality samples were selected using an active learning strategy, forming a temporal–spatial–spectral

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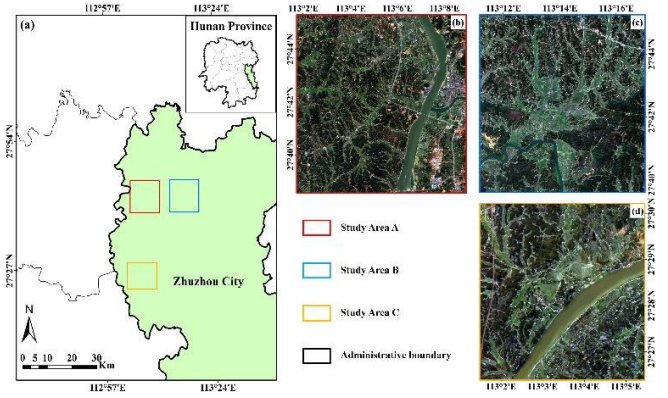


Fig. 1. Geographical location of the study area. (a) Study area is in Zhuzhou City, Hunan province. (b)–(d) Sentinel-2 images of Areas A, B, and C, respectively, acquired on March 20, 2020 (bands 4, 3, and 2 in red, green, and blue).

integrated dataset. Subsequently, a hybrid architecture, convolutional–Transformer hybrid network (CTH-Net), was proposed, combining the CNN with Transformer networks. The CNN was used to extract local spatial and spectral features, while the Transformer captured global spatial, temporal, and spectral information, improving the model's performance in complex landscapes. Furthermore, this study designed a temporal–spatial–spectral fusion (TSSF) block, which integrates temporal, spatial, and spectral features through a multilevel attention mechanism, enhancing the model's accuracy and robustness. A residual module was also introduced to address the vanishing gradient problem in deep networks, ensuring effective information propagation. Overall, this research aims to provide an accurate and efficient solution for rice extraction through feature fusion and deep learning architecture.

II. Materials

A. Study Area

The study area is situated in Zhuzhou City, northeastern Hunan province, China (27°38′–27°45′ N, 113°01′–113°17′ E), as shown in Fig. 1. It has a subtropical monsoon climate with distinct seasons. An average temperature of 16 °C–18 °C, and annual precipitation of 1400–1600 mm—conditions well-suited for agriculture. Rice is the dominant crop, cultivated in both single-cropping and double-cropping systems. Early rice is sown in March and harvested in July; late rice is sown in late May and harvested in October. Three representative study areas, A, B, and C, were selected in Zhuzhou. All areas feature complex landscapes, with diverse land-use types—agricultural fields, forests, water bodies, and nonagricultural areas—interspersed throughout. Rice cultivation is fragmented, combining single-cropping and double-cropping systems, resulting in notable temporal and spatial variability. Some agricultural fields have been abandoned, adding to the heterogeneity. Area A was chosen for model training and testing, while Areas B and C serve to assess the model's regional transferability.

B. Sentinel-2 and Sample Data

Sentinel-2, launched by the European Space Agency (ESA), is a multispectral remote sensing satellite that provides

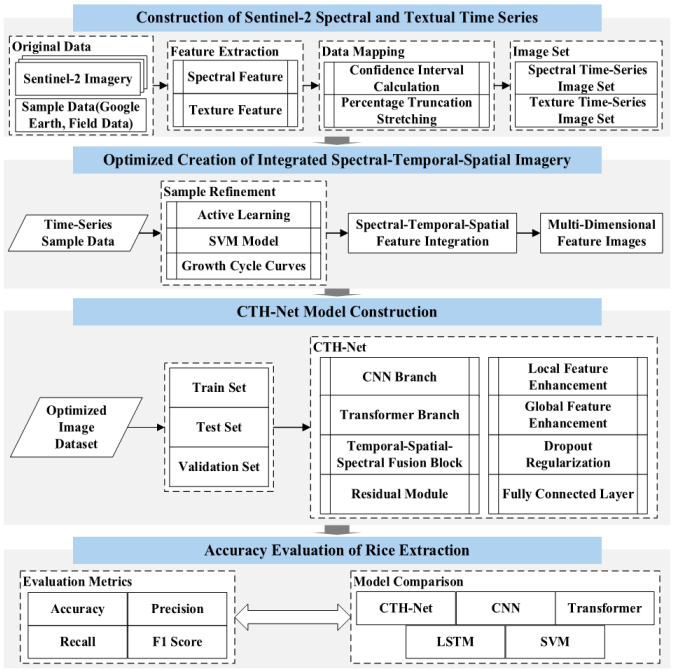


Fig. 2. Flowchart of the proposed method.

continuous, high-resolution imagery of the Earth's surface. The satellite is extensively used for land-cover classification and vegetation condition assessment. Currently, Sentinel-2A and Sentinel-2B are operational, synergistically enhancing the frequency of global observations. The multispectral imager (MSI) onboard operates at an altitude of 786 km, covering 13 spectral bands ranging from 443 to 2190 nm, with an imaging width of 290 km. Sentinel-2 offers spatial resolutions of 10, 20, and 60 m. The revisit period of a single satellite is ten days, which can be reduced to five days with the constellation of two satellites. Sentinel-2 includes three red-edge bands, which are particularly advantageous for vegetation monitoring.

The sample data were obtained from field surveys and high-resolution Google Earth imagery. The field surveys were conducted from March to October 2020, focusing on land-cover types, geographic locations, rice cropping systems, and rice growth stages. To supplement sample points, we conducted visual interpretation using Google Earth imagery and integrated the results with survey data. Samples were collected for five categories: rice, abandoned land, forest, water, and others (e.g., buildings and roads) and were evenly distributed across the study area.

III. Methods

This study introduces a novel method for pixel-level rice extraction that utilizes integrated Sentinel-2 spectral, temporal, and spatial features in combination with a lightweight hybrid deep learning architecture, named CTH-Net, to identify single-cropping and double-cropping rice systems. The workflow is depicted in Fig. 2. First, time-series images of Sentinel-2 spectral and texture features were constructed. Second, an active learning strategy was employed to selectively annotate high-confidence samples, whose temporal, spatial, and spectral

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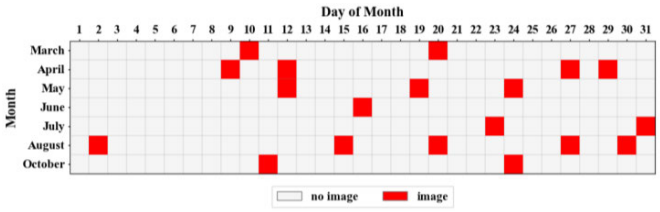


Fig. 3. Time distribution of selected Sentinel-2 images.

information was then integrated to create a unified spectral–temporal–spatial imagery dataset. Third, the CTH-Net was established for rice extraction. Finally, the accuracy of the extraction results was evaluated using a suite of assessment metrics, and a comparative analysis was conducted with the results from four other methods: CNNs, Transformers, LSTM, and SVMs.

A. Image Preprocessing

From the Copernicus Data Space Ecosystem, we downloaded 127 Sentinel-2 Level-2A image products, covering the period from March to October 2020 (https://dataspace.copernicus.eu). These images were batch-processed using the Sentinel Application Platform's graphical processing framework command-line tools to ensure consistency and efficiency. The image preprocessing involved several key steps. First, key spectral bands, including B2, B3, B4, B5, B6, B7, B8, B8A, B11, and B12, were selected to provide essential data for spectral analysis. Next, the bands with a 20-m resolution were resampled to a unified 10-m resolution to eliminate analysis biases caused by varying resolutions. Subsequently, images acquired simultaneously were mosaicked to achieve continuous coverage of the entire study area. Finally, the mosaicked images were cropped based on the precise geographic extent of the study area. Only cloud-free regions and time points were selected for analysis to ensure data quality. These processing steps ultimately produced 19 clear images (Fig. 3).

B. Construction of Sentinel-2 Spectral and Textual Time Series

This study leverages the rich spectral information provided by Sentinel-2 imagery, which spans from the visible to the shortwave infrared (SWIR) regions. The selection of ten original spectral bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, and B12) ensures the capture of comprehensive features, particularly the red-edge characteristics, which are known for their sensitivity to vegetation status and growth [70]. To better reflect vegetation vigor, water content, and soil conditions, thereby enhancing rice extraction performance, we calculated five spectral indices. The normalized difference vegetation index (NDVI) and the enhanced vegetation index (EVI) are effective in detecting vegetation changes and have been widely used for crop mapping [20], [71]. The normalized difference water index (NDWI) was employed to detect moisture content in soil and vegetation, essential for pinpointing water-dependent rice fields [8]. Additionally, the modified soil-adjusted vegetation index (MSAVI) and the brightness index 2 (BI2) account for soil background effects

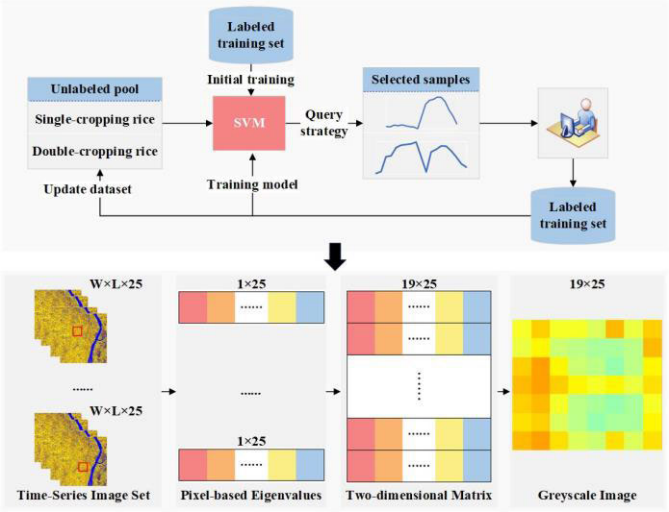


Fig. 4. Schematic of creating optimized spectral–temporal–spatial imagery.

and brightness differences across land covers, which is crucial for accurate rice extraction in scenarios with complex soil conditions or low vegetation coverage [72]. The mathematical expressions are as follows:

$$\mathrm{NDVI} = (b8 - b4)/(b8 + b4) \tag{1}$$

$$\mathrm{EVI} = 2.5 * ((b8 - b4)/(b8 + 6 * b4 - 7.5 * b2 + 1)) \tag{2}$$

$$\mathrm{NDWI} = (\mathrm{b3} - \mathrm{b8})/(\mathrm{b3} + \mathrm{b8}) \tag{3}$$

$$\mathrm{MSAVI} = \left(2 * \mathrm{b8} + 1 - \sqrt{(2 * \mathrm{b8} + 1)^2 - 8 * (\mathrm{b8} - \mathrm{b4})}\right) \Big/ 2 \tag{4}$$

$$\mathrm{BI2} = \sqrt{\left(\mathrm{b4}^2 + \mathrm{b3}^2 + \mathrm{b8}^2\right)/3} \tag{5}$$

where b2, b3, b4, and b8, respectively, represent the reflectance values of the corresponding bands. Texture features reveal local variations in pixel intensity, helping to distinguish land-cover types and reveal structural patterns [73]. The gray-level co-occurrence matrix (GLCM) captures gray-level distributions across different directions, distances, and intensities. We used GLCM to calculate the frequency of pixel value combinations and derived features such as homogeneity, angular second moment (ASM), contrast, energy, maximum probability, dissimilarity, variance, entropy, mean, and correlation to describe image texture. To optimize extraction, we applied principal component analysis (PCA) to extract the first principal component (PC1) for GLCM calculation. PCA was employed to reduce texture feature dimensionality, with the PC1 chosen to retain maximal variance. This preserves essential texture information, facilitating rice classification and reducing computational burden. The images were quantized to 32 gray levels, with pixel displacement set to 1, directions of 0°, 45°, 90°, and 135°. We averaged the texture feature values across these directions to obtain a set of comprehensive texture features for analysis. By incorporating original spectral bands and these indices as input features, we aim to enhance the model's discriminative power, enabling it to capture meaningful patterns and relationships for rice extraction.

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Given the differences in the magnitude of these feature values, we implemented a data mapping strategy to standardize them into a normalized feature space. This ensures feature comparability and enhances data processing consistency, thereby improving analysis reliability. The strategy is based on confidence intervals and percentage truncation stretching techniques. We identified a confidence interval that broadly covers the data distribution, capturing the main distribution while excluding extreme values. We selected the 1%–99% interval and calculated the grayscale values corresponding to the 1st and 99th percentiles. These values served as reference points for linear stretching, mapping the original grayscale values into a unified interval. This process enhanced pixel contrast and allowed different feature values to be represented on the same scale. Based on this, we constructed a time series by stacking 25 features for each of the 19 time points, resulting in a chronological image sequence with 25 bands per time step.

C. Optimized Creation of Integrated Spectral–Temporal–Spatial Imagery

To obtain a robust set of high-quality rice samples and enhance the accuracy of rice extraction, an active learning-based strategy was employed for the selection of single-cropping and double-cropping rice samples. Active learning is a machine learning paradigm in which a model interactively queries the most informative and uncertain samples for manual labeling, thereby maximizing learning efficiency while minimizing manual annotation effort [74], [75]. Initially, 500 field-collected single-cropping rice samples and 500 field-collected double-cropping rice samples were manually labeled based on the interpretation of NDVI time-series curves. Double-cropping rice exhibited two distinct NDVI peaks, corresponding to an early growing season (March–July) and a late growing season (July–October), while single-cropping rice displayed a single peak from late May to October. The study area is dominated by rice cultivation. Other crops have smaller planting areas and distinct growth cycles compared to rice. Therefore, differentiating single-cropping and double-cropping rice using unimodal and bimodal NDVI features is effective. These manually labeled samples were used to train an initial SVM classifier. Subsequently, the SVM model was applied to predict rice types in the remaining unlabeled samples. Samples with a predicted confidence probability exceeding 85% were considered high-confidence and directly incorporated into the sample set. Samples with lower confidence were manually reviewed and relabeled to ensure labeling accuracy. In each iteration, a subset of low-confidence samples was manually labeled and incorporated into model retraining to progressively improve the sample selection accuracy. Furthermore, to guarantee the reliability of the final dataset, even the high-confidence samples selected by SVMs were subjected to a rapid manual verification step before being incorporated into the final training set, ultimately establishing a comprehensive, high-confidence rice sample dataset.

Based on the refined samples, we constructed an integrated spectral–temporal–spatial image set to comprehensively capture the synergy of temporal, spatial, and spectral features. First, starting from the initial time point of the time series, the full band values for each pixel in the image were extracted and structured into a 1-D feature vector. This process was repeated for all subsequent time points to obtain complete temporal information. Subsequently, the 1-D feature vectors obtained from the 19 time points were integrated into a 2-D matrix, which was then converted into a 19×25-pixel grayscale image, where each pixel value reflects the spectral or spatial characteristics at a specific time point. For an image with dimensions of $W \times L$ pixels, this method is applied to each pixel, generating $W \times L$ grayscale images that collectively form a comprehensive multidimensional feature dataset. Through this approach, we created detailed temporal profiles of the spectral and spatial responses for each pixel at different time points, effectively integrating temporal, spatial, and spectral features (Fig. 4).

D. CTH-Net Architecture

The CTH-Net is designed as a hybrid architecture combining CNNs and Transformer mechanisms to enhance spectral, spatial, and temporal feature extraction for accurate rice extraction, and its architecture is shown in Fig. 5. To enhance the ability to extract higher-level abstract features of the target rice, the CNN and Transformer were employed to mine deeper representative features. The image samples are fed into the CNN branch to extract the combined features of local spatial and spectral information at various time points. The CNN branch consists of three convolutional layers, each of which is followed by batch normalization and a rectified linear unit (ReLU) activation function to ensure effective and stable feature extraction. Simultaneously, the Transformer branch is used to extract global spatial–temporal–spectral features from the input data. The convolutional layer maps the input data into a high-dimensional feature space, followed by positional encoding to preserve temporal information. The three-layer Transformer encoder captures the correlations between different time points and extracts global features of rice through the self-attention mechanism and the multilayer perceptron (MLP).

To effectively combine the local features from the CNN branch and the global features from the Transformer branch, we proposed a TSSF block with multilevel attention mechanisms. A spatial attention (SA) mechanism is incorporated to emphasize the response of key regions in the feature maps extracted by convolutional operations. This mechanism enhances detail extraction by integrating diverse local information using a combination of average and max-pooling techniques, thereby mitigating misclassification of land cover due to abandoned land phenomena and fragmented rice distribution. Its mathematical expression is as follows:

$$\begin{aligned} M_s(F) &= \sigma\left(C^{3\times3}([GMP(F(x)); GAP(F(x))])\right) \\ &= \sigma\left(C^{3\times3}\left(F^s_{\max}; F^s_{\text{avg}}\right)\right) \end{aligned} \tag{6}$$

where $F^s_{\max}$ represents the global max-pooling feature, F^s_{avg} represents the global average pooling feature, σ is the sigmoid function, and $C^{3\times3}$ denotes that the size of the convolution kernel is 3×3. A channel attention (CA) mechanism is

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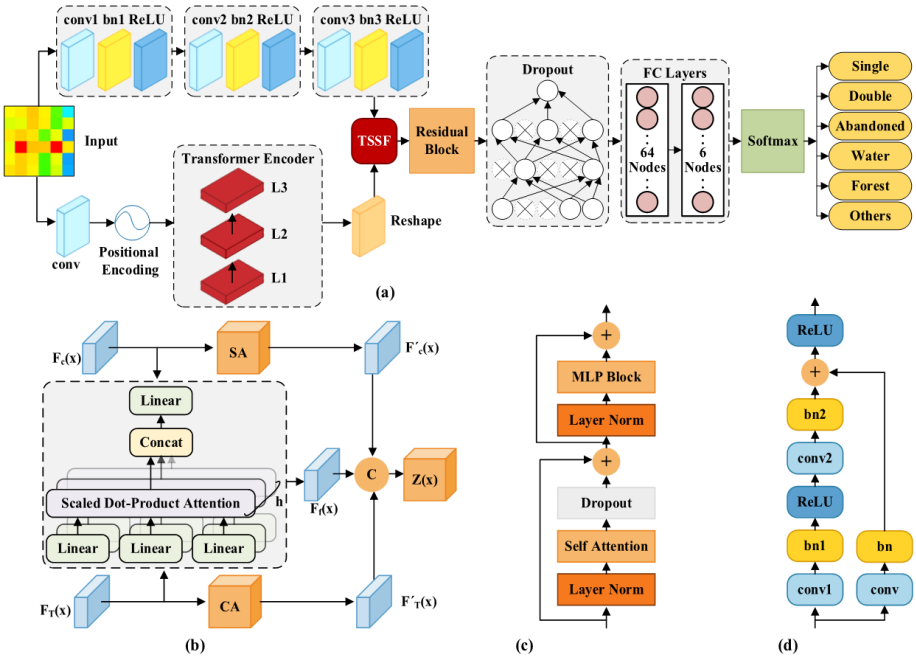


Fig. 5. Detailed structure of the CTH-Net. (a) Model's overall structure. (b) TSSF block. (c) Transformer encoder. (d) Residual module.

incorporated into the Transformer branch to adaptively adjust the weights of each channel to enhance feature representation. Adaptive max-pooling and average pooling are applied to the input features, followed by a two-stage channel feature transformation using 1-D convolution and ReLU activation, which simplifies the model while retaining cross-channel feature extraction capabilities. Unlike traditional convolution, which assigns equal weights to all channels, the CA module learns the importance of each channel and selectively enhances or suppresses specific features. This mechanism allows the model to focus on the most relevant channels, thus improving classification accuracy. Its mathematical expression is as follows:

$$S^c_{\max} = \mathrm{AMP}(S) \tag{7}$$

$$S^c_{\mathrm{avg}} = \mathrm{AAP}(S) \tag{8}$$

$$M_c(S) = \sigma\left(C^{1\times1}\left(\mathrm{ReLU}\left(C^{1\times1}\left(S^c_{\max}\right)\right)\right) + C^{1\times1}\left(\mathrm{ReLU}\left(C^{1\times1}\left(S^c_{\mathrm{avg}}\right)\right)\right)\right) \tag{9}$$

where $S^c_{\max}$ represents the adaptive max-pooling feature, S^c_{avg} represents the adaptive average pooling feature, $C^{1\times1}$ denotes that the size of the convolution kernel is 1×1, and σ is the sigmoid function. A multihead attention mechanism is utilized to fuse spatial, temporal, and spectral features extracted by the CNN and Transformer branches. Unlike traditional attention, which uses a single weight matrix, multihead attention calculates multiple attention heads in parallel, each with its own weight matrix. This allows the model to capture different aspects of the input data and learn diverse representations. Its mathematical expression is as follows:

$$\mathrm{MultiHead}(Q, K, V) = \mathrm{Concat}(\mathrm{head}_1, \mathrm{head}_2, \ldots, \mathrm{head}_h)W^O \tag{10}$$

$$\mathrm{head}_i = \mathrm{Attention}(Q_i, K_i, V_i) = \mathrm{softmax}\left(\frac{Q_i K_i^{\mathrm{T}}}{\sqrt{d_k}}\right)V_i \tag{11}$$

where head_i represents the ith attention head, Q_i, K_i, and V_i represent the three sets of query, key, and value matrices generated for each attention head, respectively, Softmax is used to normalize the attention weights, d_k denotes the feature dimension of Q, K, and V in each head, and W^O is a learnable linear transformation matrix. The outputs from all attention heads are concatenated and linearly transformed to produce the result. The outputs from the multilevel attention mechanism are merged along the channel axis, which are then processed through a residual module to yield the final feature representation.

The residual module is designed to facilitate the learning of complex feature representations while alleviating the vanishing gradient problem commonly encountered in deep networks. It consists of two convolutional layers, each followed by batch normalization to stabilize training. A 1×1 convolution shortcut connection directly links the input and output of the module, allowing the model to learn residual mappings rather than complete transformations. In this study, the residual block helps preserve temporal–spatial–spectral features during feature fusion. The shortcut connection allows effective gradient flow, enabling the model to capture complex relationships in the input data, which improves the accuracy of rice extraction under complex surface conditions [45]. Dropout regularization is applied to enhance the model's generalization ability and reduce the risk of overfitting. Finally, a fully connected layer maps the features to the category space, and a Softmax function is used to convert the output into a probability distribution, which addresses the diversity of land-cover types.

E. Model Training

The proposed CTH-Net model was implemented using the CPU version of PyTorch. During the training process,

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TABLE I
NUMBER OF SAMPLES WITHIN THE AREA A

Category	Train/Val/Test	Field Samples
Single-cropping rice	3021/378/378	500
Double-cropping rice	2107/263/263	500
Abandoned land	1281/160/160	400
Forest	4000/500/500	300
Water	3504/438/438	300
Others	4000/500/500	400

the Adam optimizer was employed to update model parameters, with an initial learning rate of 0.001. To enhance convergence and avoid overfitting, the learning rate was dynamically adjusted using the ReduceLROnPlateau strategy, which reduces the learning rate by a factor of 0.5 if the validation loss does not decrease for five consecutive epochs, with a minimum learning rate of 0.000001. The cross-entropy loss function was used to measure the difference between predicted and actual categories. A batch size of 128 was used to balance computational efficiency and generalization, and a dropout layer with a rate of 0.1 was applied to prevent overfitting and improve robustness. The model was trained for 100 epochs, and the best model was selected based on validation performance. The model was trained on a workstation equipped with an Intel Core i9 processor and 64 GB RAM. On this setup, the training time for an epoch was approximately 37 s using a batch size of 128. Training the model for 100 epochs took roughly 1 h. Given the relatively fast training time and moderate hardware requirements, CTH-Net is computationally efficient and suitable for large-area rice extraction tasks.

F. Accuracy Evaluation of Rice Extraction

To evaluate the performance of the proposed CTH-Net, all samples were divided into training, validation, and test sets in a ratio of 8:1:1. The training set was used to train the model, and the validation set was employed to monitor model convergence and guide hyperparameter adjustments. The test set was utilized for the final accuracy assessment. The quantity and spatial distribution of samples for each category in Area A are detailed in Table I and Fig. 6.

Classification performance was evaluated using a confusion matrix and four widely adopted metrics: accuracy, precision, recall, and $F1$-score. These metrics provide a comprehensive assessment of model performance across all land-cover types. Accuracy indicates the proportion of correctly classified samples among all samples. Precision measures the proportion of correctly identified samples in each predicted class. Recall reflects the proportion of actual positive samples correctly identified, and the $F1$-score is the harmonic mean of precision and recall, providing a balanced evaluation of classification performance. In addition, a comparative evaluation was conducted against four other models—CNNs, Transformers, LSTM, and SVMs—to assess the advantages of the proposed hybrid architecture. Furthermore, to evaluate the model's generalization ability, we tested its performance on independent areas (Areas B and C), providing insights into regional transferability.

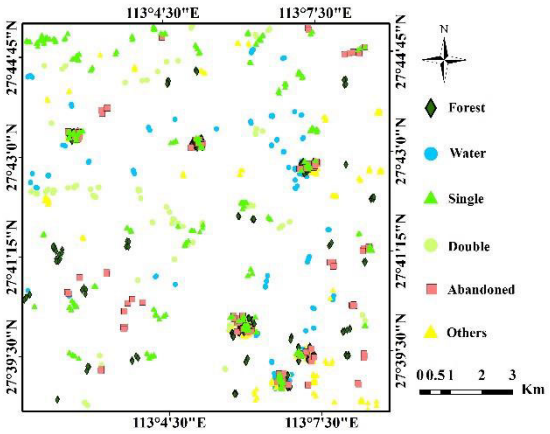


Fig. 6. Spatial distribution of field samples in Area A.

In addition to traditional accuracy metrics, we applied Shapley additive explanations (SHAP) to interpret the feature importance and understand the contribution of each input feature to the model's output. SHAP is a unified framework based on cooperative game theory that attributes the prediction of a model to its input features by estimating their marginal contributions [76]. In this study, we employed the GradientExplainer from the SHAP Python library, which is specifically designed for deep learning models and approximates Shapley values using model gradients. The input samples were passed to the trained CTH-Net model to compute SHAP values, and the mean SHAP value across these samples was used to quantify the importance of each feature.

IV. RESULTS

A. Spectral and Textual Time-Series Construction Results

Fig. 7 illustrates the temporal variations of spectral features (NDVI and NDWI) and texture features (contrast and variance) for six land-cover types (single-cropping rice, double-cropping rice, abandoned land, water bodies, forests, and others) across 19 time points, which correspond to March 10, March 20, April 9, April 12, April 27, April 29, May 12, May 19, May 24, June 16, July 23, July 31, August 2, August 15, August 20, August 27, August 30, October 11, and October 24. In Fig. 7(a), NDVI values for single-cropping and double-cropping rice exhibit a clear upward trend during the growing season, reflecting increased vegetation cover. Single-cropping rice shows one peak during its growth cycle, while double-cropping rice presents two peaks, highlighting the differences between the two cropping systems. In contrast, abandoned land shows minimal fluctuations and overall low values, indicating low vegetation cover. Forests maintain high, stable NDVI values, while water bodies remain consistently low. Fig. 7(b) shows NDWI variations. Water bodies have consistently higher NDWI values than other land types, reflecting stable water content. Single-cropping rice maintains stable NDWI values from time points 1 to 9, then declines, reaching its lowest at time point 15 before

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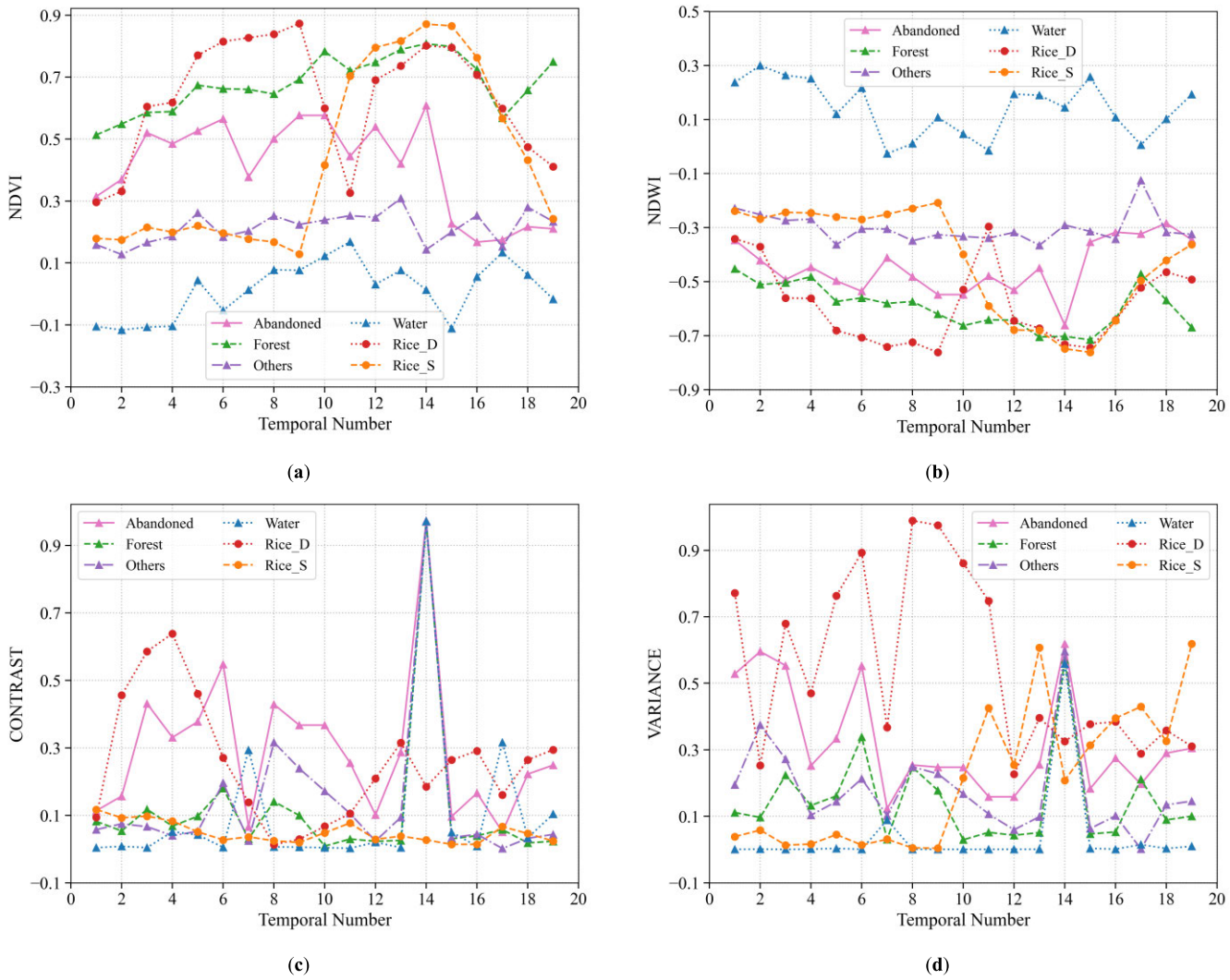


Fig. 7. Time-series curves of (a) NDVI, (b) NDWI, (c) contrast, and (d) variance for different land-cover types: abandoned land, water, forest, single-cropping rice, double-cropping rice, and others.

rising again. Double-cropping rice declines from time points 1 to 9, peaks at time point 11, and then follows a pattern similar to single-cropping rice. These fluctuations correspond to field management activities such as irrigation and drainage. Abandoned land, forests, and other land types exhibit smaller NDWI variations.

Fig. 7(c) depicts contrast variations. Single-cropping rice maintains low, stable contrast values, while double-cropping rice shows two distinct fluctuations, reflecting differences in cropping cycles and surface characteristics. The stability of single-cropping rice aligns with its uniform growth cycle and consistent management practices, while the two fluctuations observed for double-cropping rice indicate significant field changes caused by two planting cycles. Fig. 7(d) shows variance variations. Both single-cropping and double-cropping rice exhibit fluctuations during their growth cycles, indicating texture changes. Additionally, abandoned land, forests, water bodies, and other types show notable peaks at time point 14, likely due to changes in vegetation density, water conditions, or surface structures. In summary, temporal variations in spectral and texture features demonstrate differences between land types, providing a reliable basis for rice classification. Spectral indices like NDVI and NDWI effectively distinguish vegetation from nonvegetation, while texture metrics such as contrast and variance add valuable texture information to enhance classification accuracy.

B. Feature Importance of Integrated Spectral–Temporal–Spatial Imagery

Fig. 8 shows the SHAP value distributions for 25 features across 19 time points, detailing their contributions to classifying six land-cover types. Features 1–5 are NDVI, EVI, NDWI, BI2, and MSAVI. Features 6–15 correspond to the reflectance values of the spectral bands B2, B3, B4, B5, B6, B7, B8, B8A, B11, and B12. Features 16–25 include texture metrics: ASM, contrast, dissimilarity, energy, entropy, correlation, mean, variance, homogeneity, and maximum probability. For single-cropping and double-cropping rice, Features 1 and 3 show strong positive contributions during key growth stages, especially in the second growing season of double-cropping rice, aligning with the temporal trends of NDVI and NDWI in Fig. 7. Spectral features such as Features 10 and 13–15, along

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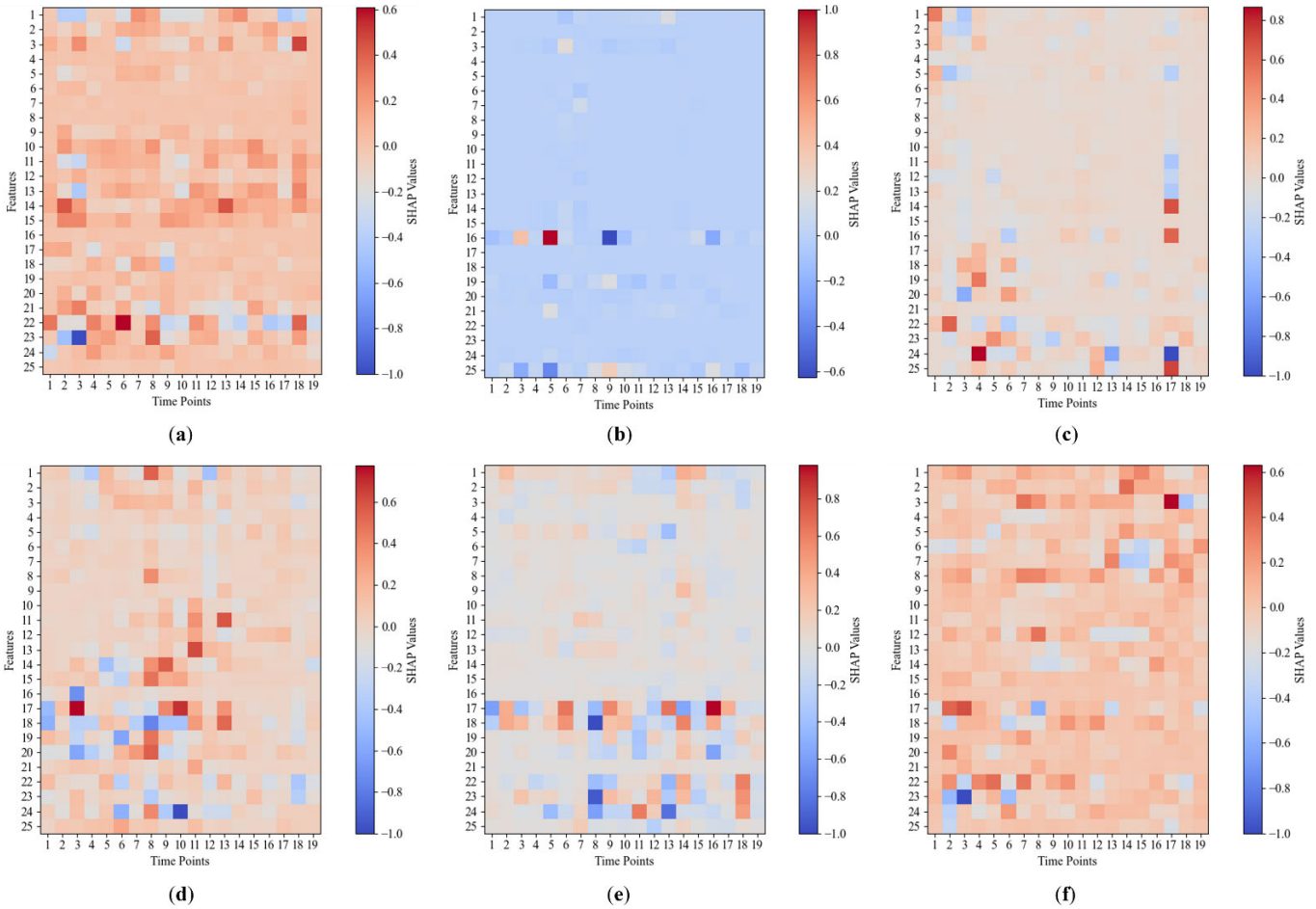


Fig. 8. SHAP-based visualization of feature contributions for (a) abandoned land, (b) water, (c) forest, (d) single-cropping rice, (e) double-cropping rice, and (f) others.

with texture features such as Features 17–20 and 24, contribute notably to single-cropping rice extraction. In contrast, double-cropping rice relies more on texture features, particularly Features 17, 18, and 22–24, reflecting the increased texture heterogeneity during its growth phases.

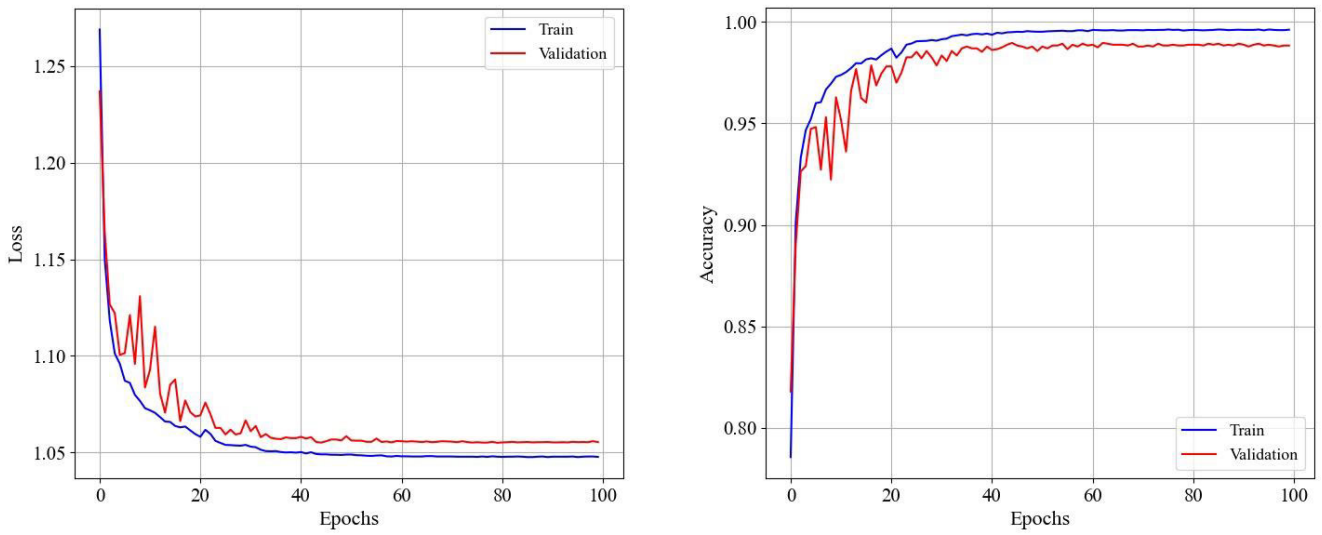
For forests, NDVI and NDWI show high contributions at certain time points, consistent with Fig. 7(a) and (b), where forests maintain consistently high NDVI values and low NDWI values, creating distinct differences from other land types. Texture features such as Features 16, 17, 19, 20, and 25 also play important roles in forest classification. For abandoned land, Features 10–15 show higher positive contributions at multiple time points, indicating their significant role in classification. Additionally, texture features such as Features 21–24 also exhibit significant contributions at certain times. For water bodies, the SHAP value analysis indicates that the contributions of the 25 features are relatively evenly distributed, showing positive contributions at most time points. NDWI exhibits a high positive contribution at time point 6, consistent with Fig. 7(b). For the "others" category, both spectral and texture features contribute to the classification of this category, but no clear patterns are observed, due to the complexity and heterogeneity of this land type. By integrating time-series curves and SHAP value analysis, the key features and contribution patterns for different land types are clearly identified. The complementary roles of spectral and texture features provide multidimensional information to enhance the accuracy of rice extraction.

C. Rice Extraction Based on the CTH-Net

Table II describes the iterative active learning process used to construct a high-quality rice sample dataset. Initially, 1000 field-collected samples of single-cropping and double-cropping rice were manually labeled to train the initial SVM model. The model then predicted 5410 candidate samples, of which 4003 high-confidence ones (probability >85%) were directly incorporated. From the remaining 1407 low-confidence samples, 200 were manually labeled to refine the model. The updated model was applied to 1207 samples, yielding 854 high-confidence and 100 manually labeled samples. A third iteration on 353 samples added 295 high-confidence samples, while the remaining 58 were manually labeled. All selected samples were ultimately verified to ensure labeling accuracy, resulting in a comprehensive, high-confidence rice sample set.

Fig. 9 illustrates the loss and accuracy trends during the training and validation processes. Both training and validation losses decreased rapidly at the start, indicating efficient

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(a) (b)

Fig. 9. Changes in (a) loss and (b) accuracy during the model training and validation process.

TABLE II
ITERATIVE ACTIVE LEARNING PROCESS FOR RICE SAMPLE SELECTION

Step	Number of samples predicted	High-confidence samples	Manually labeled samples	Notes
First prediction	5410	4003	200	Initial SVM model
Second prediction	1207	854	100	Updated SVM model
Third prediction	353	295	58	Final SVM model
Final verification	-	-	All samples manually verified	Full quality control

learning. After about 60 epochs, the losses stabilized, with minimal differences between the two sets, suggesting no overfitting. The training loss dropped from 1.2689 to approximately 1.0437, while the validation loss followed a similar trend, declining from 1.2369 to approximately 1.0559. Both training and validation accuracies increased rapidly with the number of epochs and stabilized at high levels about epoch 60. By the end, the training accuracy was maintained at around 0.9960, while the validation accuracy was sustained at around 0.9888. These results demonstrate that the model performed well on both datasets, achieving high classification accuracy.

Fig. 10(a) shows the classification map of six land-cover types, including forest, water, single-cropping rice, double-cropping rice, abandoned land, and others. The study area is predominantly covered by forests, which occupy a large portion of the region. Water bodies are primarily associated with the Xiangjiang River trunk stream, along with several scattered lakes and irrigation channels distributed around farmland. Due to the region's topography and agricultural practices, rice fields are mainly distributed in plains and valleys. The "others" category mainly includes buildings, roads, and bare land, with buildings concentrated near the central river basin. Abandoned land, influenced by both anthropogenic activities and natural factors, is sparsely scattered throughout the region and is often located within agricultural fields. Fig. 10(b) highlights the extraction results for single-cropping and double-cropping rice, two critical agricultural land types in the study area. Single-cropping rice occupies a larger area and is distributed extensively across most parts of the study region. In contrast, double-cropping rice accounts for a relatively smaller area and is more concentrated in the central parts of the study area. Additionally, double-cropping rice shows a scattered distribution in other regions and is often located near water resources. These results are consistent with our field survey findings, indicating the robustness and reliability of the CTH-Net model.

D. Rice Extraction Evaluation Results

Fig. 11 presents the land-cover classification results for six test areas using five different methods: CTH-Net, CNNs, Transformers, LSTM, and SVMs, alongside high-resolution Google Earth imagery and overlaid ground-truth labels for reference. The selected regions are characterized by complex and dispersed land-cover types, with rice fields often confused with other categories. A comparative analysis between the classification results and the reference data reveals that the CTH-Net achieves the highest consistency with the reference imagery, effectively delineating boundaries and reducing interclass misclassification. In contrast, other models struggle

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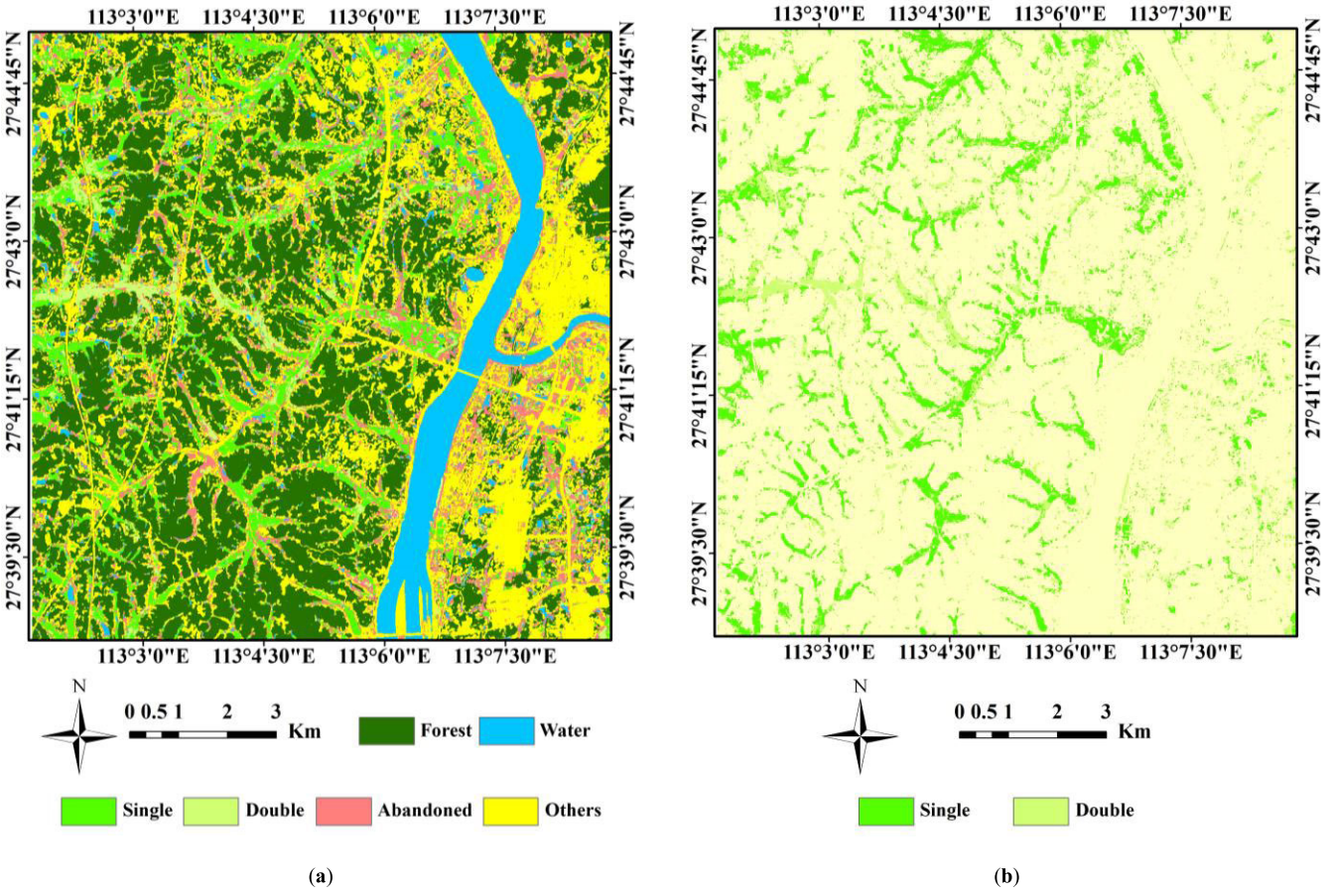


Fig. 10. Classification results of the CTH-Net. (a) Classification map of six land-cover types. (b) Rice extraction map.

with boundary clarity and interclass confusion, particularly in complex landscapes. For example, in Site 6, where water, forests, and agricultural fields are intertwined, LSTM generates blurred boundaries, and the SVM misclassifies "others" as water bodies. Similarly, in Sites 2 and 3, the Transformer, LSTM, and SVM exhibit confusion between "others" and abandoned land, due to overlapping spectral features. In Site 1, the Transformer and CNN display minor misclassifications, confusing water bodies with forests along the edges. Regarding rice fields, as seen in Sites 4, 5, and 6, the CNN, LSTM, and SVM show confusion between rice and abandoned land. By comparison, the CTH-Net's ability to integrate temporal, spatial, and spectral information enables it to effectively handle these complexities, producing results that closely align with the reference data.

The classification performance of different models in distinguishing six land-cover types is summarized in the confusion matrices (Fig. 12) and the quantitative metrics (Table III). Overall, the CTH-Net demonstrated superior performance across all categories, achieving the highest accuracy, precision, recall, and $F1$-score for most land-cover types. For double-cropping rice, the CTH-Net achieved the best results, correctly identifying 253 samples in the confusion matrix and attaining an accuracy of 0.9620. This was notably higher than the CNN (245 samples, accuracy: 0.9316), LSTM (244 samples, accuracy: 0.9278), and Transformer (235 samples, accuracy: 0.8935). Other models, particularly SVMs, exhibited greater confusion between double-cropping rice, single-cropping rice, and abandoned land. For single-cropping rice, the CTH-Net achieved a precision of 0.9812, and an $F1$-score of 0.9747, showing notable improvements over CNNs (precision: 0.9550, $F1$ score: 0.9550) and Transformers (precision: 0.9284, $F1$-score: 0.9272). For abandoned land, the CTH-Net also outperformed other models, correctly identifying 159 samples with a recall of 0.9938, followed by the Transformer (158 samples, recall: 0.9875), CNN (155 samples, recall: 0.9688), LSTM (154 samples, recall: 0.9625), and SVM (141 samples, recall: 0.8812). For water bodies, all models performed exceptionally well, indicating the effectiveness of integrating spatial, temporal, and spectral features. In forest classification, the CTH-Net maintained the highest accuracy with 500 correctly identified samples, while the CNN and SVM tied for second place (498). In the "others" category, the CNN correctly classified 500 samples, slightly outperforming the CTH-Net (499). However, the CNN misclassified more samples from single-cropping rice, double-cropping rice, forest, and abandoned land as "others," resulting in lower precision (0.9823) and $F1$-score (0.9911) compared to the CTH-Net, which achieved 0.9940 and 0.9960, respectively. These results highlight the superior accuracy of the CTH-Net, especially for single-cropping and double-cropping rice.

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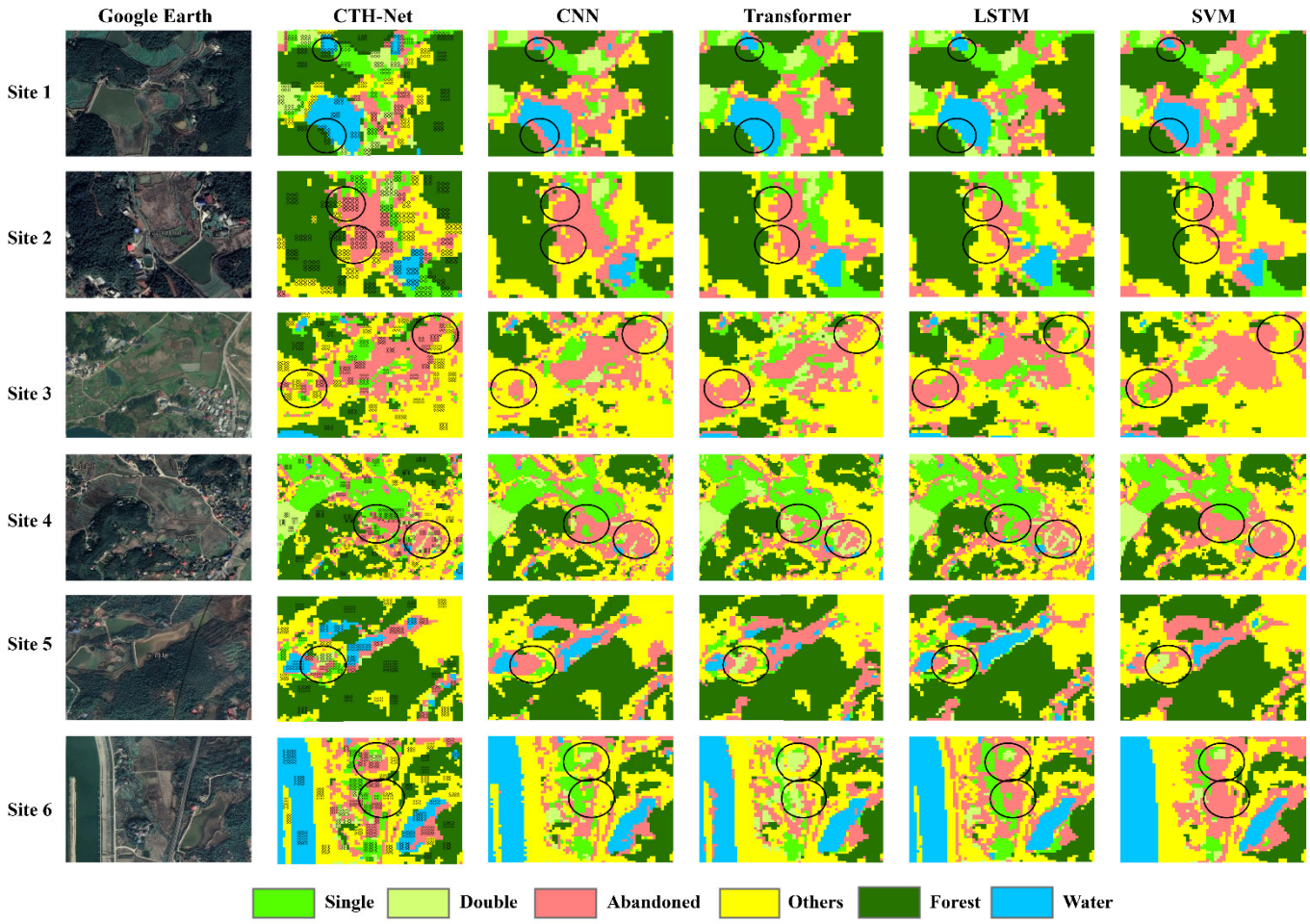


Fig. 11. Rice extraction results derived from the CTH-Net, CNN, Transformer, LSTM, and SVM in six areas.

In summary, the most frequent misclassifications occurred between single-cropping and double-cropping rice, likely due to overlapping spectral and temporal signatures during certain growth stages. Confusion also arose between rice fields and abandoned land, as both exhibit low vegetation cover and similar NDVI values during fallow periods. These errors stem from several factors: first, cloud cover reduces image availability, increases temporal gaps, and affects data continuity. Atmospheric conditions may also degrade Sentinel-2 image quality, lowering model accuracy in detecting rice growth. Second, the spectral and phenological similarities between certain classes, such as the two rice types, make them difficult to distinguish without additional contextual information.

V. Discussion

A. Ablation Experiment

To evaluate the contributions of individual components in the CTH-Net, we performed ablation experiments on four configurations: 1) baseline (CTH-Net without TSSF and residual connections); 2) baseline + TSSF; 3) baseline + residual; and 4) baseline + TSSF + residual. As shown in Table IV, the baseline model performed well overall, with an accuracy of 0.9884 and an $F1$-score of 0.9884 for the unified rice category (rice (S + D), combining single-cropping and double-cropping rice). However, the accuracies for single-cropping and double-cropping rice dropped to 0.9656 and 0.9163, and abandoned land had a lower $F1$-score of 0.9530 due to high intraclass variability. Incorporating TSSF significantly improved performance across most categories. The TSSF module integrates the CNN and Transformer architectures through a multilevel attention mechanism that enhances the model's ability to capture complex interactions across temporal, spatial, and spectral dimensions. Specifically, the SA component emphasizes informative regions within spatial feature maps, thereby improving local detail extraction by focusing on diverse and salient spatial cues. The CA module adaptively reweights feature channels by learning their relative importance, enabling the model to selectively enhance or suppress specific feature responses. This facilitates more discriminative feature representation by emphasizing channels that contribute most to the classification task. Furthermore, to effectively fuse multiscale local and global features, a multihead self-attention mechanism is employed. This mechanism aggregates complementary information from multiple attention heads, allowing the model to better exploit synergistic relationships among features at different levels. Collectively, these components strengthen the feature interaction mechanisms in the network,

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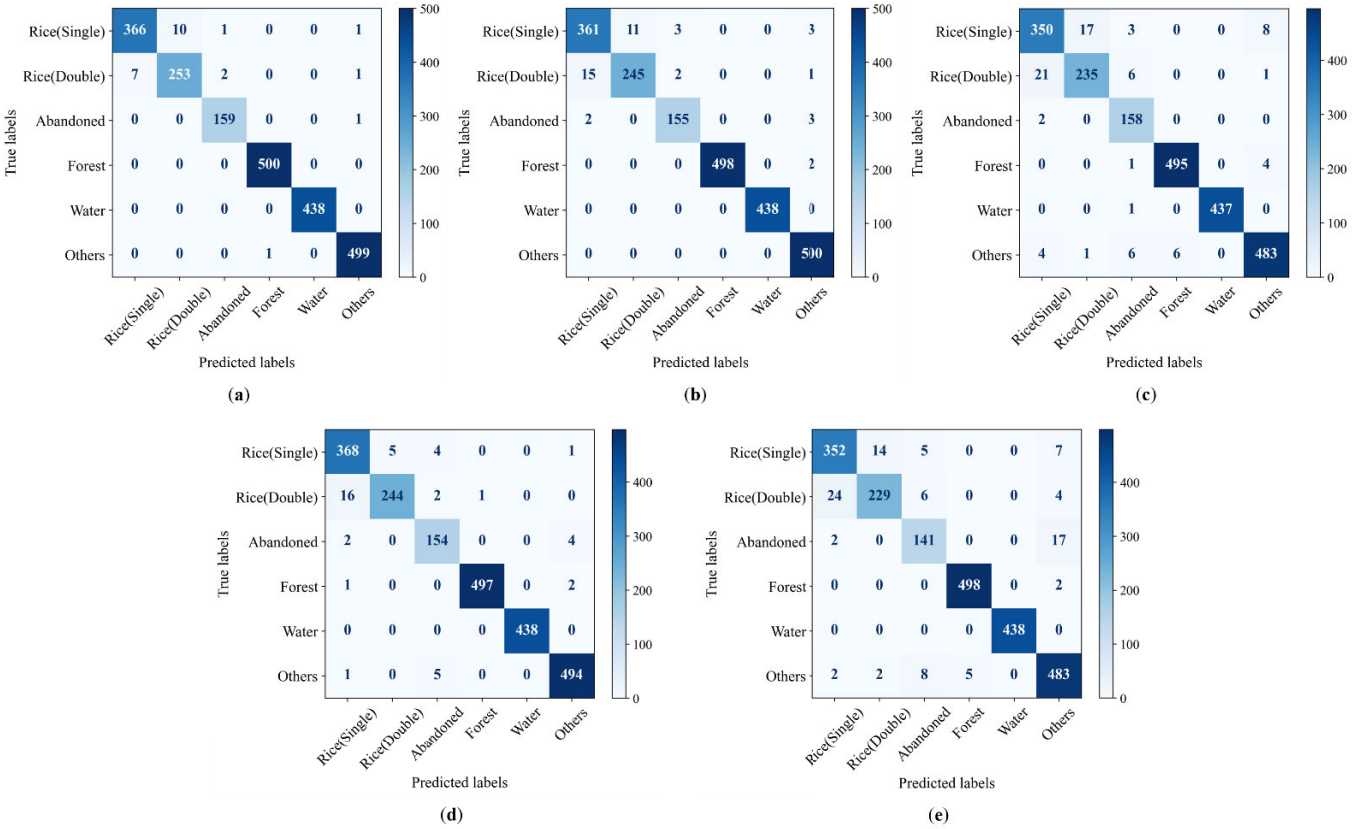


Fig. 12. Confusion matrix between predicted and actual categories based on (a) CTH-Net, (b) CNN, (c) Transformer, (d) LSTM, and (e) SVM.

enabling deeper integration of temporal, spatial, and spectral information and ultimately enhancing the model's capacity for accurate rice extraction. For the unified rice category, the accuracy increased from 0.9884 to 0.9915, and the $F1$-score rose from 0.9884 to 0.9915. For single-cropping and double-cropping rice, TSSF improved the accuracy from 0.9656 to 0.9735 and from 0.9163 to 0.9278, and the $F1$-scores to 0.9634 and 0.9476, respectively. Additionally, the precision for abandoned land increased from 0.9560 to 0.9745. These results show that TSSF enables the model to better distinguish classes with subtle differences and overlapping spectral signatures by leveraging complementary cues from multiple domains.

Residual connections also contributed to improved performance, with $F1$-scores for single-cropping and double-cropping rice increasing to 0.9658 and 0.9555, and unified rice accuracy rising to 0.9933. By introducing shortcut connections, the residual module mitigates the vanishing gradient problem, enabling the model to retain low-level spatial–spectral–temporal cues while learning high-level abstract representations. This is especially important in contexts where subtle temporal or spectral variations may be easily lost in deep layers. The residual design ensures effective information propagation, enhancing the model's ability to learn nonlinear, cross-dimensional dependencies [45]. Combining TSSF and residual connections yielded the best performance. The unified rice category achieved an accuracy of 0.9969, and the $F1$-score for double-cropping rice reached 0.9620, a substantial improvement compared to the baseline (0.9377). The precision and recall for abandoned land increased to 0.9815 and 0.9938. These gains reflect the synergistic effect of TSSF and residual connections in capturing both subtle interclass differences and maintaining high intraclass consistency. The improvements brought by combining TSSF and residual connections can be attributed to their complementary roles. TSSF strengthens multidimensional feature integration, while residual connections improve optimization and representation learning. Together, they enable the model to extract both fine-grained local details and high-level semantics, leading to more accurate rice extraction under complex landscapes.

B. Comparison With Other Methods

The comparative results in Section IV-D reveal that while all models achieved high performance on land-cover types such as water and forest, their ability to distinguish spectrally or temporally similar classes varied significantly. The CTH-Net consistently outperformed all other models, particularly in the classification of single-cropping rice ($F1 = 0.9747$), double-cropping rice ($F1 = 0.9620$), and abandoned land ($F1 = 0.9876$), due to its superior handling of the spectral and phenological similarity among these classes. In contrast, traditional models such as the SVM showed the poorest overall performance, especially for double-cropping rice ($F1 = 0.9016$). This is primarily because the SVM, being a shallow model, cannot learn complex temporal patterns or subtle spectral differences [35]. Its reliance on hand-crafted features makes it insufficient for capturing the

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TABLE III
PERFORMANCE EVALUATION OF DIFFERENT MODELS

Model	Metrics	Rice (S + D)	Single-cropping rice	Double-cropping rice	Abandoned land	Forest	Water	Others
CTH-Net	Accuracy	0.9969	0.9683	0.9620	0.9938	1.0000	1.0000	0.9980
	Precision	0.9969	0.9812	0.9620	0.9815	0.9980	1.0000	0.9940
	Recall	0.9969	0.9683	0.9620	0.9938	1.0000	1.0000	0.9980
	F1	0.9969	0.9747	0.9620	0.9876	0.9990	1.0000	0.9960
CNN	Accuracy	0.9929	0.9550	0.9316	0.9688	0.9960	1.0000	1.0000
	Precision	0.9929	0.9550	0.9570	0.9688	1.0000	1.0000	0.9823
	Recall	0.9929	0.9550	0.9316	0.9688	0.9960	1.0000	1.0000
	F1	0.9929	0.9550	0.9441	0.9688	0.9980	1.0000	0.9911
Transformer	Accuracy	0.9808	0.9259	0.8935	0.9875	0.9900	0.9977	0.9660
	Precision	0.9813	0.9284	0.9289	0.9029	0.9880	1.0000	0.9738
	Recall	0.9808	0.9259	0.8935	0.9875	0.9900	0.9977	0.9660
	F1	0.9809	0.9272	0.9109	0.9433	0.9890	0.9989	0.9699
LSTM	Accuracy	0.9897	0.9735	0.9278	0.9625	0.9940	1.0000	0.9880
	Precision	0.9899	0.9485	0.9799	0.9333	0.9980	1.0000	0.9860
	Recall	0.9897	0.9735	0.9278	0.9625	0.9940	1.0000	0.9880
	F1	0.9898	0.9608	0.9531	0.9477	0.9960	1.0000	0.9870
SVM	Accuracy	0.9732	0.9312	0.8707	0.8812	0.9960	1.0000	0.9660
	Precision	0.9735	0.9263	0.9347	0.8812	0.9901	1.0000	0.9415
	Recall	0.9732	0.9312	0.8707	0.8812	0.9960	1.0000	0.9660
	F1	0.9733	0.9288	0.9016	0.8812	0.9930	1.0000	0.9536

TABLE IV
IMPACT OF DIFFERENT BLOCKS ON THE MODEL'S PERFORMANCE

Block	Metrics	Rice (S + D)	Single-cropping rice	Double-cropping rice	Abandoned land	Forest	Water	Others
Baseline	Accuracy	0.9884	0.9656	0.9163	0.9500	0.9860	1.0000	0.9940
	Precision	0.9884	0.9456	0.9602	0.9560	0.9880	0.9955	0.9861
	Recall	0.9884	0.9656	0.9163	0.9500	0.9860	1.0000	0.9940
	F1	0.9884	0.9555	0.9377	0.9530	0.9870	0.9977	0.9900
Baseline + TSSF	Accuracy	0.9915	0.9735	0.9278	0.9563	0.9940	1.0000	0.9960
	Precision	0.9915	0.9534	0.9683	0.9745	0.9900	0.9977	0.9901
	Recall	0.9915	0.9735	0.9278	0.9563	0.9940	1.0000	0.9960
	F1	0.9915	0.9634	0.9476	0.9653	0.9920	0.9989	0.9930
Baseline + Residual	Accuracy	0.9933	0.9709	0.9392	0.9625	1.0000	1.0000	0.9960
	Precision	0.9934	0.9607	0.9724	0.9506	1.0000	1.0000	0.9901
	Recall	0.9933	0.9709	0.9392	0.9625	1.0000	1.0000	0.9960
	F1	0.9933	0.9658	0.9555	0.9565	1.0000	1.0000	0.9930
Baseline + TSSF + Residual	Accuracy	0.9969	0.9683	0.9620	0.9938	1.0000	1.0000	0.9980
	Precision	0.9969	0.9812	0.9620	0.9815	0.9980	1.0000	0.9940
	Recall	0.9969	0.9683	0.9620	0.9938	1.0000	1.0000	0.9980
	F1	0.9969	0.9747	0.9620	0.9876	0.9990	1.0000	0.9960

nuanced growth cycles that differentiate single-cropping and double-cropping rice [36]. The significant performance gap between the SVM and deep learning models suggests that modeling spatiotemporal continuity is crucial for accurate classification in heterogeneous agricultural landscapes. While LSTM performed better than the SVM in handling temporal

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TABLE V
COMPARISON WITH PREVIOUS STUDIES ON RICE EXTRACTION

Category	Dataset	Model	Reported Metric			Reference
			Rice(D+S)	Single-cropping rice	Double-cropping rice	
Advanced deep learning for rice extraction	Sentinel-2	A-BiGRU	F1(0.98)	-	-	[77]
	Sentinel-2	CTANet	F1(0.96)	-	-	[78]
	Sentinel-1	ESF-Seg	Accuracy(0.98), F1(0.95)	-	-	[79]
	Sentinel-2	ConvLSTM2D	Accuracy(0.97)	-	-	[30]
Identification of single-cropping and double-cropping rice	Sentinel-2	Optimal temporal windows	-	F1(0.94)	-	[80]
	Sentinel-1/2	Red-edge index RERI	-	F1(0.92)	F1(0.93)	[81]
	Sentinel-2	Phenology index RPI	-	F1(0.92)	-	[82]
General machine/deep learning for rice classification	Sentinel-2	RF	Accuracy(0.96)	-	-	[83]
	Sentinel-2	3D-CNN	Accuracy(0.96)	-	-	[84]
This study	Sentinel-2	CTH-Net	Accuracy(0.99), F1(0.99)	Accuracy(0.97), F1(0.97)	Accuracy(0.96), F1(0.96)	-

sequences, they still underperformed the CTH-Net due to their limited spatial modeling ability. LSTM achieved high recall for single-cropping rice (0.9735) but lower recall for double-cropping rice (0.9278), indicating that it could detect the rice sequence but often misclassified subtle intraclass variations—likely due to its sequential processing structure and lack of local spatial awareness [55], [56], [57].

CNNs demonstrated strong performance on land-cover types with distinct spatial boundaries or textures such as water, forest, and other categories, owing to their strength in extracting local features. However, they struggled with double-cropping rice ($F1 = 0.9441$) because their limited receptive field prevents them from capturing long-range dependencies in temporal or spectral dimensions, which are essential for differentiating rice cropping patterns over time [31], [47]. Transformers, with their global attention mechanisms, addressed some of these limitations by modeling long-range temporal dependencies [58], [59], [60]. However, their performance on double-cropping rice was still lower than that of the CTH-Net. This can be attributed to the Transformer's lack of strong inductive biases for local spatial structure, which makes it less effective in learning fine-grained spatial patterns crucial for distinguishing small or adjacent rice parcels [62]. The CTH-Net mitigated these issues by synergistically combining a CNN branch (for local spatial and spectral extraction) with a Transformer branch (for global temporal modeling), and by employing TSSF and residual modules to highlight the most discriminative features. This hybrid structure enabled a balanced high performance across all categories, including abandoned land, single-cropping, and double-cropping rice.

To further demonstrate the superiority of the proposed method, we compared it with three categories of previous studies: 1) recent deep learning models for rice extraction; 2) studies distinguishing single-cropping and double-cropping rice; and 3) general machine/deep learning models (Table V). Recent deep learning methods, such as Transformer-based or ConvLSTM-based architectures, have demonstrated strong performance in rice classification tasks. However, most of these approaches focus only on binary classification (rice versus nonrice), without considering the finer distinctions between cropping systems. On the other hand, studies that specifically classify single-cropping and double-cropping rice often rely on traditional models such as RF. While these methods incorporate phenological or temporal features, they often lack the deep feature representation capabilities of modern neural networks. Compared with previous studies, the CTH-Net not only achieves more precise rice extraction, distinguishing between single-cropping and double-cropping rice but also balances performance with computational efficiency, making it suitable for large-scale applications.

C. Regional Transferability of the CTH-Net

As shown in Fig. 13, the regional transferability of the proposed CTH-Net model was evaluated in Areas B and C. In both regions, forests are the dominant land-cover type, extensively spread throughout the study areas. Agricultural lands also occupy a large proportion, primarily distributed across plains and valleys. Water bodies are centered around the Xiangjiang River and its tributaries, supplemented by

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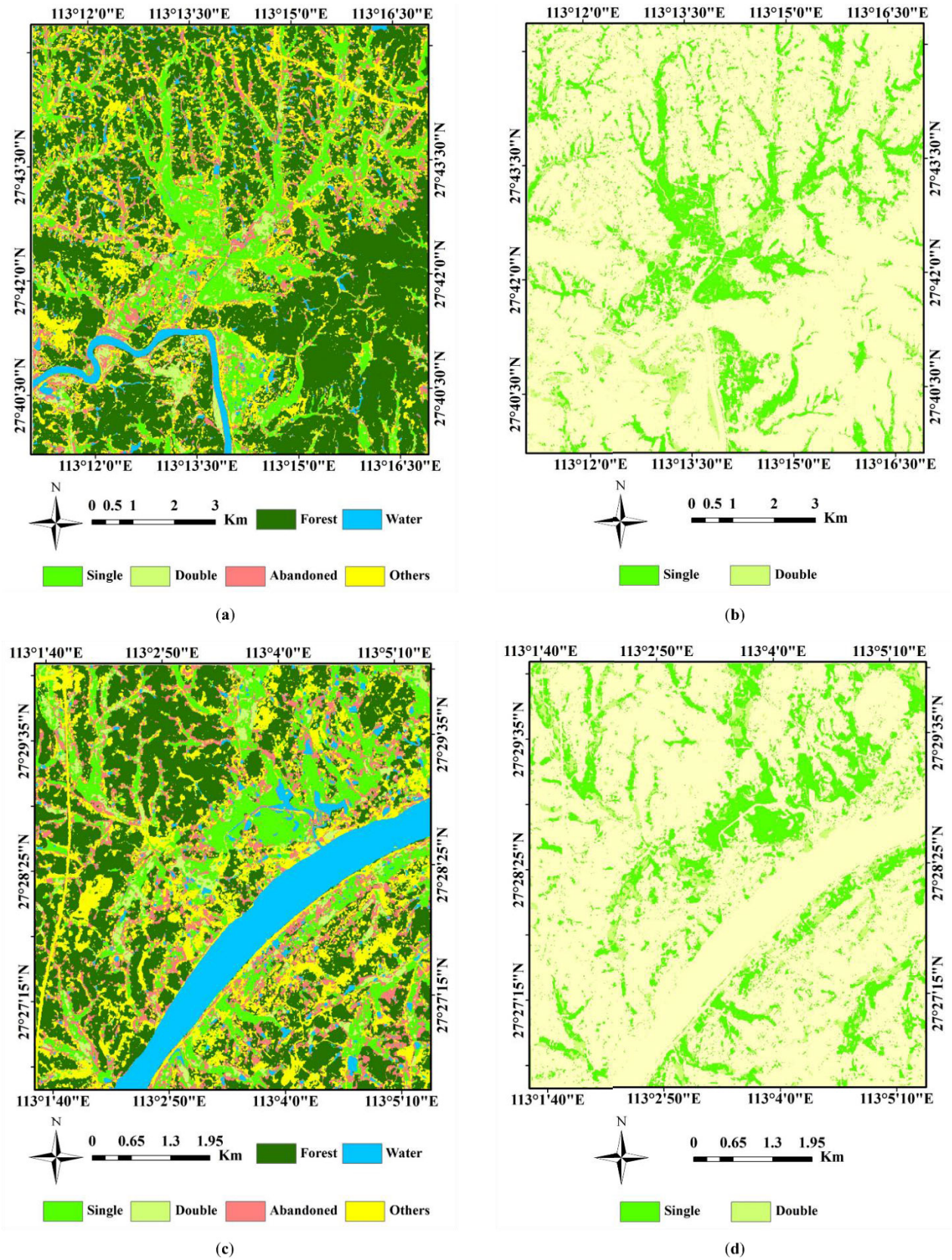


Fig. 13. Classification results of the CTH-Net in the model transfer area. (a) and (c) Classification maps of six land-cover types for Areas B and C. (b) and (d) Rice extraction maps for Areas B and C.

scattered lakes and irrigation channels that support agricultural activities. Other land types, such as buildings and roads, are scattered around forests and farmland. Abandoned land is predominantly located within agricultural fields

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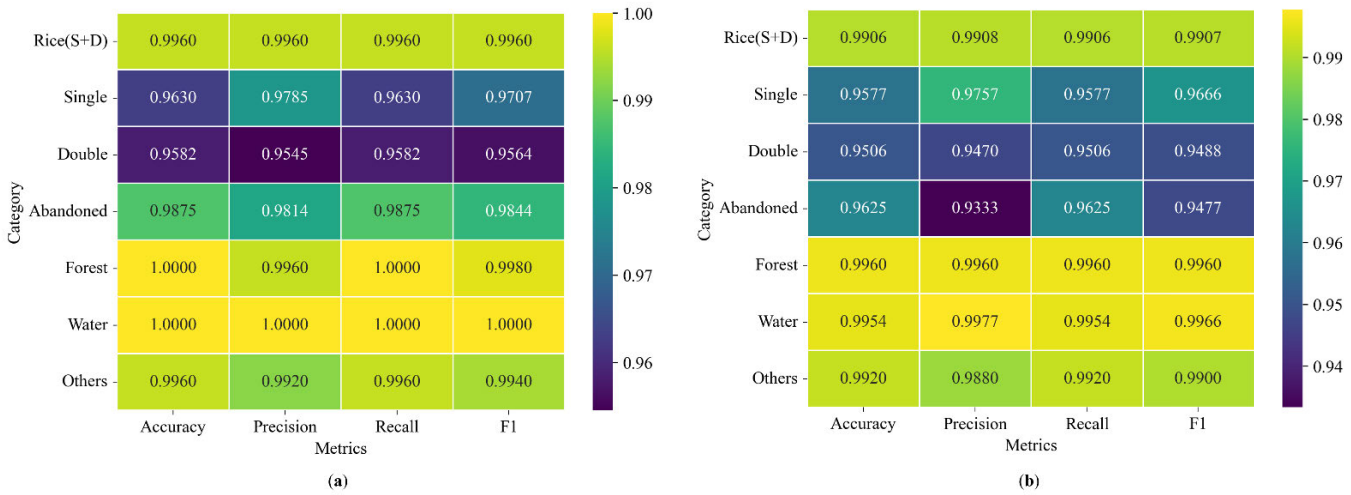


Fig. 14. CTH-Net's regional transferability evaluation for (a) Area B and (b) Area C.

or adjacent to water bodies. In terms of rice cropping systems, single-cropping rice occupies a larger area than double-cropping rice in both regions. It is widely distributed across most parts of the study areas, whereas double-cropping rice is mainly concentrated in the central and southern regions or near major water sources, reflecting its high dependence on sufficient irrigation. This spatial distribution aligns well with local water resource management and irrigation practices.

Fig. 14 illustrates the transferability evaluation of the CTH-Net across six land-cover types in transfer regions B and C. The evaluation was conducted using field survey samples, with the same number of class-wise samples as in Area A. Overall, the model demonstrates strong generalization across regions. For rice extraction, the CTH-Net not only accurately identifies rice-growing areas but also effectively distinguishes between single-cropping and double-cropping rice. This indicates its ability to capture the temporal patterns of different rice cultivation modes. The combined rice category achieves consistently high scores across all four metrics, further confirming the model's robustness and adaptability in cross-regional rice extraction. Besides rice, the model also performs excellently on natural land-cover types like forests and water bodies. Although the classification performance for abandoned land and the "others" category shows slight fluctuations, the overall accuracy remains high, suggesting that the model maintains good adaptability in complex land-cover environments.

D. Advantages and Limitations

In complex landscapes, rice extraction is challenged by diverse land-use types, spatiotemporal variability, planting system differences, and fragmented rice distribution. Moreover, the presence of abandoned land further complicates land-cover types, negatively impacting classification accuracy. Therefore, it is crucial to propose methods that can effectively address these issues. First, this study presents a rice extraction method based on spectral and texture feature time series. Qu et al. [78] demonstrated that temporal changes in spectral features can effectively differentiate crops across seasons. Additionally, texture features, as noted by Feng et al. [85], complement spectral information and improve classification accuracy for complex land covers. Using Sentinel-2 spectral and texture time series, we captured the temporal variation of land-cover types, including single-cropping rice, double-cropping rice, abandoned land, water bodies, forests, and others. The temporal patterns of NDVI and NDWI reflect rice growth cycles and field management activities, while texture features such as contrast and variance highlight spatial differences between land types, offering valuable cues for accurate rice classification. Second, we designed an optimized method for creating integrated spectral–temporal–spatial imagery. Active learning strategies were employed to automatically select high-confidence samples, reducing manual labeling efforts while enhancing the quality and diversity of the training dataset. By combining spectral, temporal, and spatial features and fully considering their spatiotemporal correlations, we built a high-dimensional feature set that comprehensively captures rice growth dynamics at various time points.

Third, we proposed a hybrid deep learning architecture combining the CNN and Transformer, termed the CTH-Net. This architecture extracts local spatial and spectral features at various time points using the CNN, while the Transformer captures global spatiotemporal–spectral features. We introduced a TSSF block with multilevel attention mechanisms, which integrates SA, CA, and multihead attention mechanisms. This enables the model to focus on specific spatiotemporal regions and feature channels from multiple angles and levels, optimizing feature representation. Compared to other models (CNNs, Transformers, LSTM, and SVMs), this mechanism reduces the misclassification of land covers, particularly when distinguishing rice from other types. Moreover, the residual module alleviates gradient vanishing issues in deep networks, ensuring effective feature learning during model training. Ablation studies further confirm that both TSSF and residual connections significantly improve model performance, with the best performance achieved when both components are integrated.

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Previous studies have shown the effectiveness of attention mechanisms and residual networks in improving classification accuracy, which is consistent with our results [77], [86], [87]. With this design, the CTH-Net better captures seasonal and spatial variations in rice, especially under complex conditions, significantly improving extraction accuracy. Results demonstrate that the CTH-Net outperforms other models in classifying all land-cover types, with clear advantages in handling fragmented rice and mixed land types. To further enhance the interpretability of the model, we employed SHAP analysis to uncover the contribution of different features to rice extraction. The results revealed that spectral features like NDVI and NDWI are crucial during key growth stages of both single-cropping and double-cropping rice, especially in the second growing season of double-cropping rice. NDVI reflects vegetation greenness and can be used as an indicator of crop vigor and biomass accumulation [20], while NDWI captures water content in vegetation and soil, which is particularly relevant for water-demanding crops like rice [8]. Texture features, such as contrast, dissimilarity, and homogeneity, were crucial to both single-cropping and double-cropping rice, consistent with the temporal trends presented in Fig. 7, further validating the effectiveness of the proposed method. Notably, some features exhibited negative SHAP values for certain land-cover classes. This does not imply these features are unhelpful for the classification process. Rather, a negative SHAP value indicates that the current feature reduces the likelihood of assigning the sample to that specific class. However, the same feature may positively contribute to other classes, enhancing the model's overall discrimination. These negative SHAP values can mainly be attributed to two mechanisms: first, the current feature deviates from the typical pattern of that class, prompting the model to lower its classification confidence; and second, in multiclass settings, classification relies on relative differences in feature distributions, so features that suppress one class may promote another, thus serving as key indicators for interclass differentiation [76]. For instance, for abandoned land, NDVI at time point 2 and variance at time point 3 displayed negative SHAP values, suggesting their suppressive roles. Yet, these features showed positive contributions for single-cropping and double-cropping rice, aiding class-specific identification. Such features that serve contrasting roles across classes are essential for improving the model's ability to distinguish target categories. Additionally, the CTH-Net model exhibited robust regional transferability, rapid training convergence, and moderate computational demands. This suggests that the CTH-Net not only excels in the study area but is also highly generalizable for rice extraction in diverse regions.

Despite this study has yielded promising results, several limitations need to be considered for further improvement. First, remote sensing imagery can be affected by cloud cover and atmospheric conditions, which may degrade data quality at certain time points and impact the model's accuracy in assessing rice growth. Although data preprocessing and normalization techniques were applied in this study, the model's robustness to outliers in complex environments still needs improvement. Therefore, advanced atmospheric correction algorithms should be developed to enhance reflectance accuracy. Second, while an active learning strategy was employed to enhance dataset quality, manual labeling remains inefficient. Future work could focus on reducing human intervention by developing more efficient automated sample selection methods. For example, incorporating generative adversarial networks (GANs) to generate synthetic data could enhance model adaptability in diverse conditions. Third, while the CTH-Net performed well in Areas B and C, its transferability to broader regions and varying agricultural environments requires further validation. Differences in land use, cropping systems, and management practices may affect the model's performance. Therefore, it is necessary to conduct additional validation in different regions in the future to ensure the stability and reliability of the model under varying conditions.

VI. Conclusion

This study presents a novel approach for rice extraction in complex landscapes by integrating Sentinel-2 spectral–temporal–spatial data with a hybrid deep learning model, CTH-Net. The combination of spectral, temporal, and spatial information, along with the introduction of a TSSF block and a residual module, enhances the model's ability to accurately extract rice in complex landscapes. The results show that the CTH-Net achieved 99.69% accuracy, precision, recall, and $F1$-score in rice extraction, consistently performing well across different categories, including single-cropping rice, double-cropping rice, and abandoned land. The CTH-Net outperformed other models like CNNs, Transformers, LSTM, and SVMs in distinguishing rice from other land types. The feature importance analysis revealed that integrating spectral, temporal, and spatial information is essential for accurate rice extraction. The model adapted its feature utilization according to the land type: for single-cropping rice, spectral and spatial features at specific time points were more influential, whereas, for double-cropping rice, textural features at particular growth stages played a more significant role. Ablation experiments confirmed the contributions of the TSSF block and the residual module in improving model performance. The inclusion of both modules significantly enhanced accuracy and robustness, validating the effectiveness of these architectural enhancements. Furthermore, the CTH-Net demonstrates the potential for regional transferability, showing adaptability to different agricultural environments. This study provides an efficient and reliable solution for rice extraction in complex landscapes, offering a robust tool for dynamic rice monitoring in areas with high spatiotemporal heterogeneity.

References

[1] S. Muthayya, J. D. Sugimoto, S. Montgomery, and G. F. Maberly, "An overview of global Rice production, supply, trade, and consumption," *Ann. New York Acad. Sci.*, vol. 1324, no. 1, pp. 7–14, Sep. 2014.
[2] N. K. Fukagawa and L. H. Ziska, "Rice: Importance for global nutrition," *J. Nutritional Sci. Vitaminol.*, vol. 65, pp. S2–S3, Oct. 2019.
[3] N. Poole, J. Donovan, and O. Erenstein, "Viewpoint: Agri-nutrition research: Revisiting the contribution of maize and wheat to human nutrition and health," *Food Policy*, vol. 100, Apr. 2021, Art. no. 101976.
[4] H. Zhang, H. Yuan, W. Du, and X. Lyu, "Crop identification based on multi-temporal active and passive remote sensing images," *ISPRS Int. J. Geo-Inf.*, vol. 11, no. 7, p. 388, Jul. 2022.

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[5] F. Keppler, J. T. G. Hamilton, M. Braß, and T. Röckmann, "Methane emissions from terrestrial plants under aerobic conditions," *Nature*, vol. 439, no. 7073, pp. 187–191, Jan. 2006.

[6] B. A. Linquist, M. A. Adviento-Borbe, C. M. Pittelkow, C. van Kessel, and K. J. van Groenigen, "Fertilizer management practices and greenhouse gas emissions from Rice systems: A quantitative review and analysis," *Field Crops Res.*, vol. 135, pp. 10–21, Aug. 2012.

[7] A. D. Smartt, K. R. Brye, and R. J. Norman, "Methane emissions from Rice production in the United States—A review of controlling factors and summary of research," *Greenhouse Gases*, vol. 30, no. 30, pp. 179–207, Mar. 2016.

[8] J. Dong and X. Xiao, "Evolution of regional to global paddy Rice mapping methods: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 119, pp. 214–227, Sep. 2016.

[9] E. R. Hunt and C. S. T. Daughtry, "What good are unmanned aircraft systems for agricultural remote sensing and precision agriculture?" *Int. J. Remote Sens.*, vol. 39, nos. 15–16, pp. 5345–5376, Aug. 2018.

[10] M. E. D. Chaves, M. C. A. Picoli, and I. D. Sanches, "Recent applications of Landsat 8/OLI and Sentinel-2/MSI for land use and land cover mapping: A systematic review," *Remote Sens.*, vol. 12, no. 18, p. 3062, Sep. 2020.

[11] M. Weiss, F. Jacob, and G. Duveiller, "Remote sensing for agricultural applications: A meta-review," *Remote Sens. Environ.*, vol. 236, Jan. 2020, Art. no. 111402.

[12] M. Mosleh, Q. Hassan, and E. Chowdhury, "Application of remote sensors in mapping Rice area and forecasting its production: A review," *Sensors*, vol. 15, no. 1, pp. 769–791, Jan. 2015.

[13] R. Zhao, Y. Li, and M. Ma, "Mapping paddy Rice with satellite remote sensing: A review," *Sustainability*, vol. 13, no. 2, p. 503, Jan. 2021.

[14] J. Wang et al., "Mapping paddy Rice planting area in wheat-rice double-cropped areas through integration of Landsat-8 OLI, MODIS and PALSAR images," *Sci. Rep.*, vol. 5, no. 1, May 2015, Art. no. 10088.

[15] R. Ni et al., "An enhanced pixel-based phenological feature for accurate paddy Rice mapping with Sentinel-2 imagery in Google Earth Engine," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 282–296, Aug. 2021.

[16] Y. Di, M. Gao, F. Feng, Q. Li, and H. Zhang, "A new framework for winter wheat yield prediction integrating deep learning and Bayesian optimization," *Agronomy*, vol. 12, no. 12, p. 3194, Dec. 2022.

[17] Y. Zhai, N. Wang, L. Zhang, L. Hao, and C. Hao, "Automatic crop classification in northeastern China by improved nonlinear dimensionality reduction for satellite image time series," *Remote Sens.*, vol. 12, no. 17, p. 2726, Aug. 2020.

[18] L. Yin, N. You, G. Zhang, J. Huang, and J. Dong, "Optimizing feature selection of individual crop types for improved crop mapping," *Remote Sens.*, vol. 12, no. 1, p. 162, Jan. 2020.

[19] N. Wang, Y. Zhai, and L. Zhang, "Automatic cotton mapping using time series of Sentinel-2 images," *Remote Sens.*, vol. 13, no. 7, p. 1355, Apr. 2021.

[20] X. Guan, C. Huang, G. Liu, X. Meng, and Q. Liu, "Mapping Rice cropping systems in Vietnam using an NDVI-based time-series similarity measurement based on DTW distance," *Remote Sens.*, vol. 8, no. 1, p. 19, Jan. 2016.

[21] S. Li et al., "A new method for winter wheat mapping based on spectral reconstruction technology," *Remote Sens.*, vol. 13, no. 9, p. 1810, May 2021.

[22] W. S. Moola, W. Bijker, M. Belgiu, and M. Li, "Vegetable mapping using fuzzy classification of dynamic time warping distances from time series of Sentinel-1A images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 102, Oct. 2021, Art. no. 102405.

[23] Y. Guo, X. Jia, D. Paull, and J. A. Benediktsson, "Nomination-favoured opinion pool for optical-SAR-synergistic Rice mapping in face of weakened flooding signals," *ISPRS J. Photogramm. Remote Sens.*, vol. 155, pp. 187–205, Sep. 2019.

[24] P. Zhan, W. Zhu, and N. Li, "An automated Rice mapping method based on flooding signals in synthetic aperture radar time series," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112112.

[25] J. Liao et al., "A Rice mapping method based on time-series Landsat data for the extraction of growth period characteristics," *Sustainability*, vol. 10, no. 7, p. 2570, Jul. 2018.

[26] J. Yang et al., "An automated sample generation method by integrating phenology domain optical-SAR features in Rice cropping pattern mapping," *Remote Sens. Environ.*, vol. 314, Dec. 2024, Art. no. 114387.

[27] Q. Shao et al., "Large-scale mapping of new mixed Rice cropping patterns in southern China with phenology-based algorithm and MODIS dataset," *Paddy Water Environ.*, vol. 21, no. 2, pp. 243–261, Feb. 2023.

[28] Z. Guo, K. Yang, C. Liu, X. Lu, L. Cheng, and M. Li, "Mapping national-scale croplands in Pakistan by combining dynamic time warping algorithm and density-based spatial clustering of applications with noise," *Remote Sens.*, vol. 12, no. 21, p. 3644, Nov. 2020.

[29] S. Yang, L. Gu, X. Li, F. Gao, and T. Jiang, "Fully automated classification method for crops based on spatiotemporal deep-learning fusion technology," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5405016.

[30] Q. Li, J. Tian, and Q. Tian, "Deep learning application for crop classification via multi-temporal remote sensing images," *Agriculture*, vol. 13, no. 4, p. 906, Apr. 2023.

[31] T. Nampally, K. Kumar, S. Chatterjee, R. Pachamuthu, B. Naik, and U. B. Desai, "StressNet: A spatial–spectral–temporal deformable attention-based framework for water stress classification in maize," *Frontiers Plant Sci.*, vol. 14, Nov. 2023, Art. no. 1241921.

[32] H. T. T. Nguyen et al., "Mapping maize cropping patterns in Dak Lak, Vietnam through MODIS EVI time series," *Agronomy*, vol. 10, no. 4, p. 478, Apr. 2020.

[33] F. Xuan et al., "Mapping crop type in northeast China during 2013–2021 using automatic sampling and tile-based image classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 117, Mar. 2023, Art. no. 103178.

[34] D. Mishra et al., "Crop classification by using dual-pol SAR vegetation indices derived from Sentinel-1 SAR-C data," *Environ. Monitor. Assessment*, vol. 195, no. 1, Jan. 2023, Art. no. 115.

[35] R. Interdonato, D. Ienco, R. Gaetano, and K. Ose, "DuPLO: A dual view point deep learning architecture for time series classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 149, pp. 91–104, Mar. 2019.

[36] J. Xu et al., "DeepCropMapping: A multi-temporal deep learning approach with improved spatial generalizability for dynamic corn and soybean mapping," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111946.

[37] J. Xu et al., "Towards interpreting multi-temporal deep learning models in crop mapping," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112599.

[38] L. Xia et al., "A full resolution deep learning network for paddy Rice mapping using Landsat data," *ISPRS J. Photogramm. Remote Sens.*, vol. 194, pp. 91–107, Dec. 2022.

[39] M. Rußwurm, N. Courty, R. Emonet, S. Lefèvre, D. Tuia, and R. Tavenard, "End-to-end learned early classification of time series for in-season crop type mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 445–456, Feb. 2023.

[40] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[41] H. Wang, C. Tao, J. Qi, R. Xiao, and H. Li, "Avoiding negative transfer for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4413215.

[42] R. Ahila Priyadharshini, S. Arivazhagan, M. Arun, and A. Mirnalini, "Maize leaf disease classification using deep convolutional neural networks," *Neural Comput. Appl.*, vol. 31, no. 12, pp. 8887–8895, Dec. 2019.

[43] W. X. Bao, T. Cheng, X. G. Zhou, W. Guo, and X. Zhang, "An improved dense net model to classify the damage caused by cotton aphid," *Comput. Electron. Agricult.*, vol. 203, Dec. 2022, Art. no. 107485.

[44] S. Mohammadi, M. Belgiu, and A. Stein, "Improvement in crop mapping from satellite image time series by effectively supervising deep neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 272–283, Apr. 2023.

[45] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[46] M. A. Haq, G. Rahaman, P. Baral, and A. Ghosh, "Deep learning based supervised image classification using UAV images for forest areas classification," *J. Indian Soc. Remote Sens.*, vol. 49, no. 3, pp. 601–606, Mar. 2021.

[47] F. Dao, Y. Zeng, and J. Qian, "Fault diagnosis of hydro-turbine via the incorporation of Bayesian algorithm optimized CNN-LSTM neural network," *Energy*, vol. 290, Mar. 2024, Art. no. 130326.

[48] I. A. L. Magalhães et al., "Comparing machine and deep learning methods for the phenology-based classification of land cover types in the Amazon biome using Sentinel-1 time series," *Remote Sens.*, vol. 14, no. 19, p. 4858, Sep. 2022.

[49] Z. Sun, L. Di, and H. Fang, "Using long short-term memory recurrent neural network in land cover classification on Landsat and cropland data layer time series," *Int. J. Remote Sens.*, vol. 40, no. 2, pp. 593–614, Oct. 2018.

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[50] R. N. Masolele et al., "Spatial and temporal deep learning methods for deriving land-use following deforestation: A pan-tropical case study using Landsat time series," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112600.

[51] K. Greff, R. K. Srivastava, J. Koutník, B. R. Steunebrink, and J. Schmidhuber, "LSTM: A search space Odyssey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 10, pp. 2222–2232, Oct. 2017.

[52] H. Crisóstomo de Castro Filho et al., "Rice crop detection using LSTM, bi-LSTM, and machine learning models from Sentinel-1 time series," *Remote Sens.*, vol. 12, no. 16, p. 2655, Aug. 2020.

[53] Y. Zhou, J. Luo, L. Feng, Y. Yang, Y. Chen, and W. Wu, "Long-short-term-memory-based crop classification using high-resolution optical images and multi-temporal SAR data," *GIScience Remote Sens.*, vol. 56, no. 8, pp. 1170–1191, Jul. 2019.

[54] A. Gafurov, S. Mukharamova, A. Saveliev, and O. Yermolaev, "Advancing agricultural crop recognition: The application of LSTM networks and spatial generalization in satellite data analysis," *Agriculture*, vol. 13, no. 9, p. 1672, Aug. 2023.

[55] L. Zhong, L. Hu, and H. Zhou, "Deep learning based multi-temporal crop classification," *Remote Sens. Environ.*, vol. 221, pp. 430–443, Feb. 2019.

[56] A.-X. Zhu, F.-H. Zhao, H.-B. Pan, and J.-Z. Liu, "Mapping Rice paddy distribution using remote sensing by coupling deep learning with phenological characteristics," *Remote Sens.*, vol. 13, no. 7, p. 1360, Apr. 2021.

[57] P. Wei, D. Chai, T. Lin, C. Tang, M. Du, and J. Huang, "Large-scale Rice mapping under different years based on time-series Sentinel-1 images using deep semantic segmentation model," *ISPRS J. Photogramm. Remote Sens.*, vol. 174, pp. 198–214, Apr. 2021.

[58] C. Zhang, J. Su, Y. Ju, K.-M. Lam, and Q. Wang, "Efficient inductive vision transformer for oriented object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5616320.

[59] K. Ni, Q. Wu, S. Li, Z. Zheng, and P. Wang, "Remote sensing scene classification via second-order differentiable token transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5626915.

[60] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[61] Y. Yuan and L. Lin, "Self-supervised pretraining of transformers for satellite image time series classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 474–487, 2021.

[62] Y. Chen, D. Su, and J. Luo, "Laplacian-guided hierarchical transformer: A network for medical image segmentation," *Comput. Methods Programs Biomed.*, vol. 260, Mar. 2025, Art. no. 108526.

[63] B. Niu, Q. Feng, B. Chen, C. Ou, Y. Liu, and J. Yang, "HSI-TransUNet: A transformer based semantic segmentation model for crop mapping from UAV hyperspectral imagery," *Comput. Electron. Agricult.*, vol. 201, Oct. 2022, Art. no. 107297.

[64] B. M. Varma, N. N. Naik, K. Chandrasekaran, M. Venkatesan, and J. Rajan, "Forecasting land-use and land-cover change using hybrid CNN–LSTM model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[65] P. Tang et al., "Deep learning with multi-scale temporal hybrid structure for robust crop mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 117–132, Mar. 2024.

[66] Z. Li, G. Chen, and T. Zhang, "A CNN-transformer hybrid approach for crop classification using multitemporal multisensor images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 847–858, 2020.

[67] P. Hao, L. Di, C. Zhang, and L. Guo, "Transfer learning for crop classification with cropland data layer data (CDL) as training samples," *Sci. Total Environ.*, vol. 733, Sep. 2020, Art. no. 138869.

[68] C. Zhang, H. Zhang, and L. Zhang, "Spatial domain bridge transfer: An automated paddy Rice mapping method with no training data required and decreased image inputs for the large cloudy area," *Comput. Electron. Agricult.*, vol. 181, Feb. 2021, Art. no. 105978.

[69] H. Zhang, W. Liu, and L. Zhang, "Seamless and automated rapeseed mapping for large cloudy regions using time-series optical satellite imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 45–62, Feb. 2022.

[70] Y. Sun, Q. Qin, H. Ren, T. Zhang, and S. Chen, "Red-edge band vegetation indices for leaf area index estimation from Sentinel-2/MSI imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 826–840, Feb. 2020.

[71] X. Xiao et al., "Mapping paddy Rice agriculture in southern China using multi-temporal MODIS images," *Remote Sens. Environ.*, vol. 95, no. 4, pp. 480–492, Apr. 2005.

[72] G. Rondeaux, M. Steven, and F. Baret, "Optimization of soil-adjusted vegetation indices," *Remote Sens. Environ.*, vol. 55, no. 2, pp. 95–107, Feb. 1996.

[73] H. Zakeri, F. Yamazaki, and W. Liu, "Texture analysis and land cover classification of Tehran using polarimetric synthetic aperture radar imagery," *Appl. Sci.*, vol. 7, no. 5, p. 452, Apr. 2017.

[74] B. Settles, "Active learning literature survey," Dept. Computer Sciences, Univ. Wisconsin, Madison, WI, USA, Tech. Rep. 1648, 2010.

[75] D. Tuia, E. Pasolli, and W. J. Emery, "Using active learning to adapt remote sensing image classifiers," *Remote Sens. Environ.*, vol. 115, no. 9, pp. 2232–2242, Sep. 2011.

[76] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 4768–4777.

[77] F. Feng, M. Gao, R. Liu, S. Yao, and G. Yang, "A deep learning framework for crop mapping with reconstructed Sentinel-2 time series images," *Comput. Electron. Agricult.*, vol. 213, Oct. 2023, Art. no. 108227.

[78] T. Qu et al., "A fine crop classification model based on multitemporal Sentinel-2 images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 134, Nov. 2024, Art. no. 104172.

[79] Y. Xie, L. Xu, H. Zhang, M. Song, J. Ge, and F. Wu, "Tropical Rice mapping using time-series SAR images and ESF-Seg model in Hainan, China, from 2019 to 2023," *Remote Sens.*, vol. 17, no. 2, p. 209, Jan. 2025.

[80] L. Sheng, Y. Lv, Z. Ren, H. Zhou, and X. Deng, "Detection of the optimal temporal windows for mapping paddy Rice under a double-cropping system using Sentinel-2 imagery," *Remote Sens.*, vol. 17, no. 1, p. 57, Dec. 2024.

[81] Y. Wan, Y. Gong, F. Xu, W. Shi, and W. Gao, "A novel red-edge vegetable index for paddy Rice mapping based on Sentinel-1/2 and GF-6 images," *Int. J. Digit. Earth*, vol. 17, no. 1, Sep. 2024, Art. no. 2398068.

[82] M. Yang, B. Guo, and J. Wang, "A novel and robust method for large-scale single-season Rice mapping based on phenology and statistical data," *ISPRS J. Photogramm. Remote Sens.*, vol. 213, pp. 14–32, Jul. 2024.

[83] P. Wei et al., "Early crop mapping based on Sentinel-2 time-series data and the random forest algorithm," *Remote Sens.*, vol. 15, no. 13, p. 3212, Jun. 2023.

[84] I. Gallo, L. Ranghetti, N. Landro, R. La Grassa, and M. Boschetti, "In-season and dynamic crop mapping using 3D convolution neural networks and Sentinel-2 time series," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 335–352, Jan. 2023.

[85] S. Feng, J. Zhao, T. Liu, H. Zhang, Z. Zhang, and X. Guo, "Crop type identification and mapping using machine learning algorithms and Sentinel-2 time series data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3295–3306, Sep. 2019.

[86] Z. Cai et al., "Learning spectral–spatial representations from VHR images for fine-scale crop type mapping: A case study of rice-crayfish field extraction in South China," *ISPRS J. Photogramm. Remote Sens.*, vol. 199, pp. 28–39, May 2023.

[87] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.



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